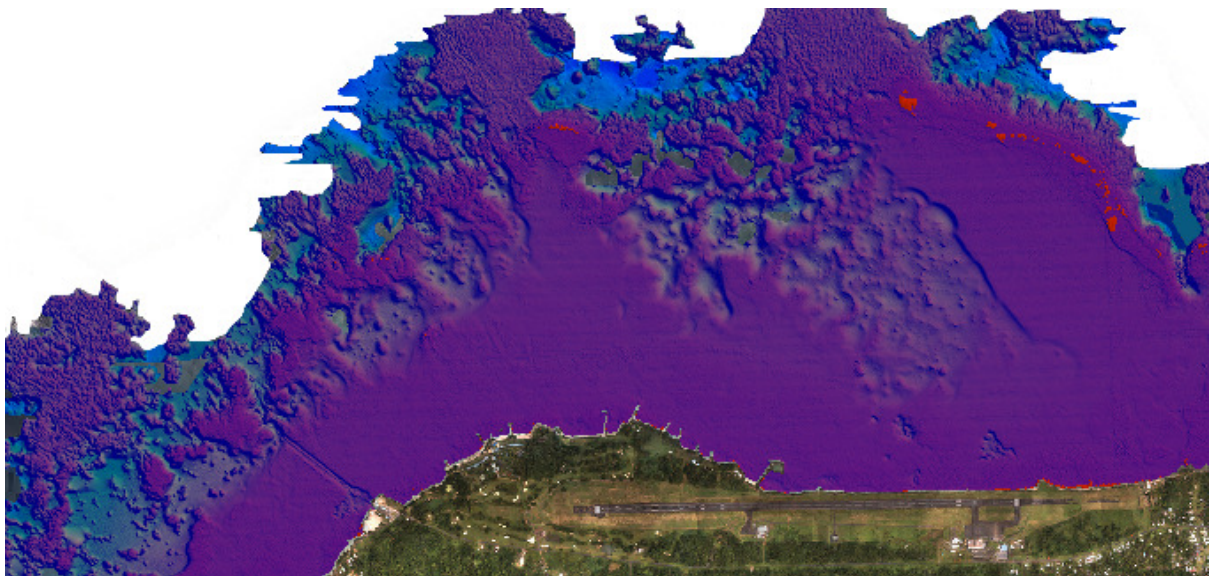




Acquisition of LiDAR and Imagery Derived Products for Samoa

December 2012

Report of Survey



Contract Number
CMC G4140

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Glossary

AOC	Ahab Operator Console
CSS	Coastal Survey Studio
EGM	Earth Gravitational Model
GA	GeoScience Australia
GCA	Ground Control Area
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GRS	Geodetic Reference System
IMU	Inertial Measurement Unit
ITRF	International Reference Frame
LADS	Laser Airborne Depth Survey
MSL	Mean Sea Level
PPK	Post Processed Kinematic
SBET	Smoothed Best Estimate of Trajectory
SCA	Sea Control Area
TIP	Topographic Integration Point
TPU	Total Propagated Uncertainty

1. Executive Summary

1.1 Overview

The Laser Airborne Depth Survey (LADS) for GeoScience Australia commenced on the 02nd December 2012 and data gathering operations were completed on the 12th December 2012. The survey was flown by Pelydryn Limited using a HawkEye IIb Bathymetric LiDAR sensor, fitted to a NOMAD aircraft provided by AAM Pty Ltd.



Figure 1: NOMAD Aircraft used for flying operations in Samoa.

Weather conditions were extremely variable with heavy rainfall experienced in showers or for prolonged periods. This had several impacts on operations, low lying cloud and rainfall prevented operations due to operational limitations and safety considerations. The rainfall experienced also caused significant run off of sediments into the coastal area and the resultant turbidity impacted on the quality of the data. The local orographic weather effects sometimes meant that flights could be continued at one end of the island, whilst raining at the other. In any case the flying programme was extended as a result of the short sorties undertaken whilst trying to progress operations between weather fronts. Sun glint was also a problem as it meant that operations could only be flown before 10am and after 2pm meaning the amount of daylight was also limited. As a result of the weather conditions there were occasions where sorties took place during the time where sun glint was a factor and if necessary lines were reflown to capture data. The impact on data quality is discussed more fully in Section 4.

Shipping and fishing activity had no impact on acquisition operations. The small numbers of vessels encountered in the port area were manually removed from the dataset during data processing.

1.2 Survey Areas

The survey area in Samoa was divided into 2 main blocks as shown in the graphics below (Figure 2);

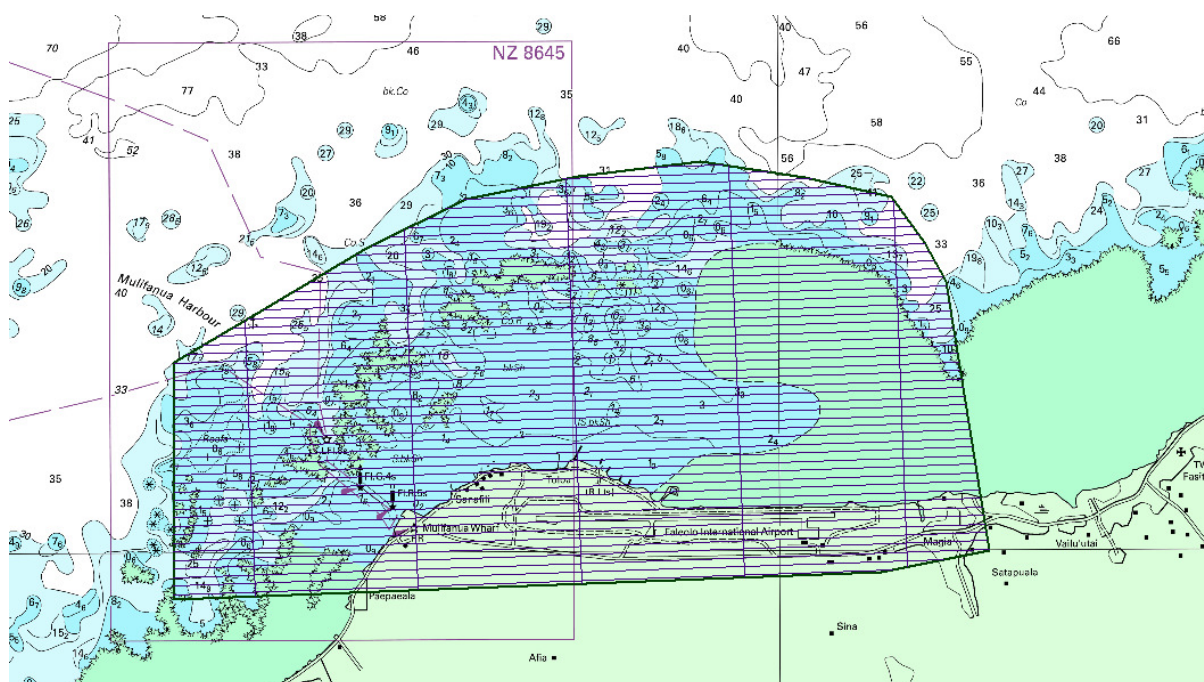


Figure 2: Survey Area Overview Area A

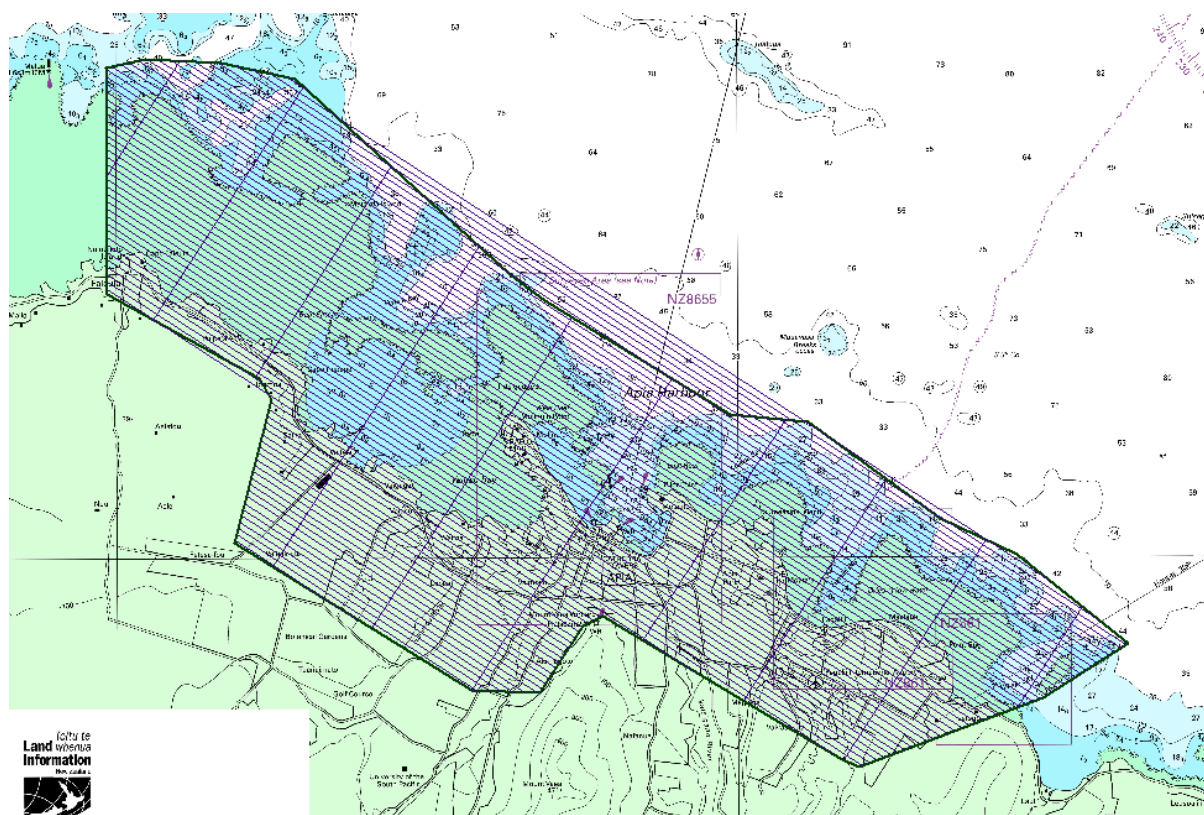


Figure 3: Survey Area Overview Area B

Areas and line directions were chosen in order to achieve a balance between data collection and

turns whilst flying.

The HawkEye IIb system was mounted on board an AAM provided NOMAD aircraft. This aircraft is well suited to Bathymetric LiDAR work, offering a stable platform and allowing the operator full access to the system in flight. The sensor installation was approved by a licensed Aircraft Engineer and all aircraft operations complied with Civil Aviation Safety Authorities (CASA) regulations at all times.

Occupational Health and Safety controls were established in the Project Safety Management Plan and were adhered to throughout survey operations. There were no reportable incidents and no injuries to personnel, damage to property or the environment during survey operations.

1.3 Survey Data

During flight operations, a number of other activities were undertaken in support of the survey. Ground and Sea Control Areas (GCA and SCA) were established by Pelydryn and AAM Pty Ltd. The system arrived in Samoa, factory and field calibrated having been installed and calibrated in Vanuatu but the new areas were used to validate the system remained in calibration.

The survey area covered a substantial topographic component as well as bathymetry to the 30m contour or polygon limits as defined in the Statement of Requirement. No additional flying was undertaken outside of these polygons. Bathymetric, topographic, position, attitude and imagery data were collected simultaneously and logged using the Operator Console software. Offline data processing was conducted using Coastal Survey Studio (Version 2.3.0.11), CARIS HIPS (Version 7.1 SP 2 HF 6) and Fledermaus (Version 7.3.2a). All of the systems generally performed well, though the large volumes of topographic data acquired did impact on the offline data processing speed particularly when processing in the extreme shallow water on the coastal fringe where the bathymetric and topographic data overlapped.

The area surveyed varied considerably. The topography consisted of areas of open grassland, coconut plantations, heavy forest and urban areas. The bathymetry was characterised by extensive shallow reef plates fringing the coastline with steep drops and coral outcrops lying offshore. Sandy beaches and rocky outcrops interspersed these areas. Where possible lines along the Low Tide Line were flown at low water and high water to allow capture across the area where waves were breaking.

Mobilisation and Acceptance Tests were completed in Brisbane and calibration flights were undertaken in Efate and are fully detailed in the Mobilisation Report (see Annex C).

1.4 Geodetics

Online positioning was provided by an Applanix POSAV. A Post Processed Kinematic (PPK) solution was utilised and applied to derive a full 3D navigation solution using Applanix POSMMS (v5.4). This uses position and attitude data logged whilst online, RINEX data from a local base station and rapid ephemeris data to derive a final navigation solution. This tightly coupled navigation solution was later applied to the raw waveform data in Coastal Survey Studio. All data is reported in the ITRF 2008 Reference frame as the AUSPOS Processing Service was used to provide base station data co-ordinates.

Dynamic and Static validation of the navigation solution were undertaken daily and as part of the initial calibration flights. Further details can be found in the Mobilisation Report (see Annex C).

1.5 Tidal Data

The Statement of Requirement required all data to be produced on the ellipsoid and reduced to Mean Sea Level (1973). Tide gauge observations were undertaken to establish Mean Sea Level (2012) at Apia as a comparison to MSL (1973). An existing tidal station was recovered at the port in Apia. Mean Sea Levels were derived and compared against MSL 1973 and EGM2008 in order to derive the accuracy of the existing EGM2008 model (see below table).

Station	Ellipsoidal Height	EGM2008 Separation Value	MSL 2012 Separation Value	EGM2008 to MSL 2012 Difference	MSL 1973 Separation Value	EGM2008 to MSL 1973 Difference
PEL2	40.049	37.144	37.915	+0.771	37.812	+0.668

1.6 Seabed Features and Contacts

The area surveyed consisted of 2 areas on the North side of the island. No specific wreck investigations were required by the contract.

1.7 Conclusion

Following extensive surveying and analysis of cleaned data in accordance with stated acceptance criteria, it is assessed that the requirement to detect all depths shoaler than 30m has been met where the seabed was detected. Data gaps do exist in the reef structures and on the seabed where no data was obtained for which there are a number of reasons:

- a. In shallow waters, the system's capability to detect both the sea surface and seabed return is highly sensitive to other factors such as glint, breaking waves, contrast and slope. This is discussed more fully at Section 3.6
- b. Turbidity – plankton and the high rainfall experienced led to significant reductions in water clarity due to run-off from shore.

The sea areas surveyed and encompassed within the limits provided with this report are surveyed to IHO Order 1B. Topographic data collected meets the requirements laid down in Section 11 (General Topographic LiDAR Specifications) of the Statement of Requirement.

2 Narrative

The system was mobilised to Brisbane on 21st September 2012. Following a delay due to customs issues and aircraft availability the system was installed in the aircraft on 18th October 2012. After local liaison was completed, operations commenced on 31st October 2012 in Vanuatu and finished on 12th December 2012 in Samoa.

2.1 Survey Activity Summary

Days on Task	Mobilisation	Surveying	Weather	Maintenance	Transit	Breakdown	Standby
11	0	10	1	0	0	0	0

Table 1: Survey Activity Summary

A Daily Narrative highlighting key milestones and events during the survey is detailed below. For further detailed daily reports see the relevant Daily Progress Reports (delivered daily during operations) and attached at Annex B.

September 2012	
UTC Date	
27 Sep	System 002 Arrives Brisbane – Held up in Customs due to licensing issues

November 2012	
26 Nov	Survey Operations Completed in Vanuatu

December 2012	
2 Dec	Field Team Arrive Samoa Field Observations
3 Dec	Survey (3.5hrs)
4 Dec	Survey (5hrs)
5 Dec	Survey (4.7hrs)
6 Dec	Standby - Weather
7 Dec	Survey (4.1hrs)
8 Dec	Survey (5hrs)
9 Dec	Survey (3.4hrs) Tide Station Observations at Apia Harbour
10 Dec	Survey (3hrs) Tide Station Observations and Levelling at Apia Harbour
11 Dec	GPS Observations at Apia Harbour Apia SCA Survey
12 Dec	Apia SCA Survey Survey Operations Completed in Samoa

3 Geodesy

3.1 Horizontal Datum, Spheroid, Projection and Grid Details

All control is referred to ITRF 2008 datum and GRS 80 spheroid, UTM Zone 2 (S) Projection.

Geodetic Reference System	
Reference Frame/Datum	ITRF 2008
Spheroid	GRS 80
Semi- Major Axis	6378137.00
Inverse Flattening	298.257222101

Projection System Apia, Samoa	
Projection	Transverse Mercator
Grid Details	UTM Grid Zone 2 South
Central Meridian (CM)	171° 00' 00" W
Latitude of Origin	000° 00' 00"N
False Easting	500,000.000
False Northing	10000000
Scale Factor on CM	0.9996

Table 2: Datum, Spheroid and Projection Details

3.2 Horizontal and Vertical Positioning

Positional control for the aircraft and installed survey system was provided by the integrated Applanix POS AV 410 Inertial Measurement Unit which is part of the Hawkeye IIb system. The survey was conducted on the ITRF 2008 datum with all outputs being related to the ellipsoid.

POS AV uses an IMU and GNSS receiver as its primary navigation sensors during flight. The embedded software runs an inertial algorithm that calculates velocity, angular movement and heading to derive an inertial position. At the same time a Kalman filter compares this position with the one provided by the GNSS receiver. The output inertial position is adjusted by this filter to remove the inertial errors that can build up. During flight operations the output position has a horizontal accuracy of between 1-3m and a vertical accuracy of 0.5-1.0m. In order to improve the horizontal and vertical accuracy of the navigation solution, Pelydryn post processes the recorded position and orientation solution to derive trajectories with accuracies better than 0.1m in all planes.

In order to gain the most accurate position and orientation solution, Pelydryn made use of a of previously co-ordinated GPS Base station and precise ephemeris to post process the aircraft navigation sensor data in Applanix's POSPAC MMS (Version 5.4) to produce a post processed navigation solution. Base Station observables were observed daily from a previously established co-ordinated Station as detailed in Table 3 below:

Base Station Name		
Samoa Faliolo Airport (OLO1)	Latitude	13° 49' 55.37420" S
	Longitude	171° 59' 53.89288" W
	Altitude	4.331m

Table 3: Base Station Co-Ordinates

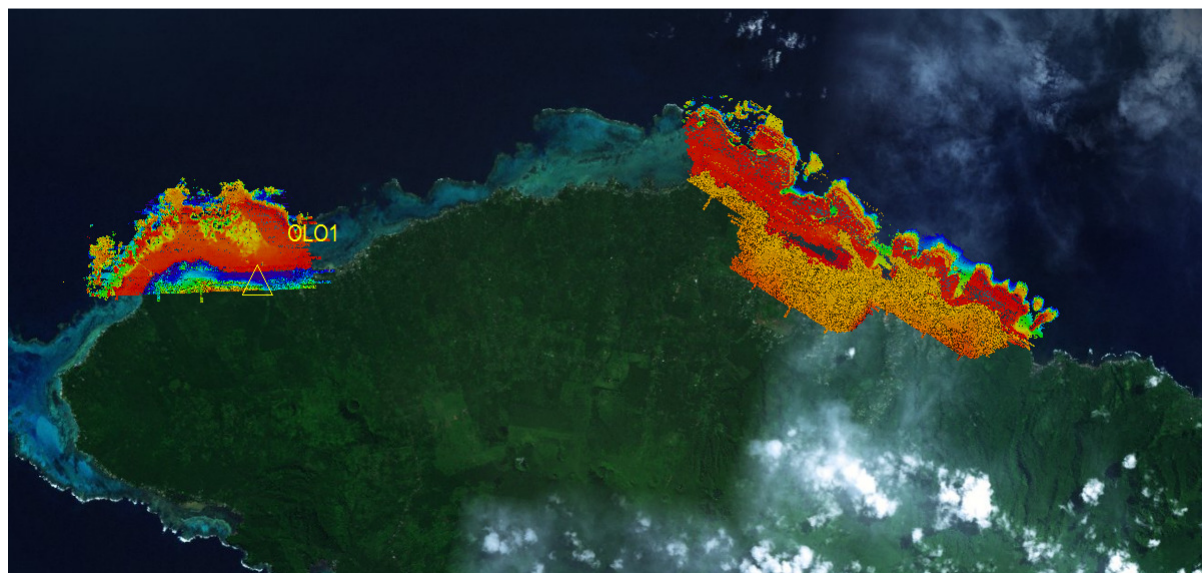


Figure 4: Position of GNSS Base Stations Apia Samoa.

Short baseline algorithms were used within PosPAC MMS to allow a single base station solution to be used. This algorithm is automatically selected within the software when the first and last 5 minutes of the mission are within 20 km of the selected base station and ensures the navigation files for this project area were within the accuracy specification. Each SBET file was reviewed to provide an error check. The POSPAC MMS Accuracy Report from each sortie is provided in Annex D.

3.3 Geodetic Observations

As part of the survey, Pelydryn also undertook a number of geodetic observations to establish GPS Benchmarks adjacent to the tidal station at Apia. Observations were undertaken using a Trimble R8 GNSS Receiver. All data was processed using the AUSPOS Processing service using final ephemeris and the control networks were adjusted by AAM using Leica Geo Office to derive final co-ordinates.

Station	Horizontal Accuracy (m)	Vertical Accuracy (m)	ITRF 2008 GRS 80 Latitude	ITRF 2008 GRS 80 Longitude	Ellipsoidal Height (m)	UTM Easting 2 S	UTM Northing 2 S	Orthometric Height (m) (EGM 2008)
OLO1	0.001	0.004	13° 49' 55.37420" S	171° 59' 53.89288" W	42.191m	392109.717	8470623.449	4.331m

PEL 1	0.001	0.006	13° 49' 35.44544" S	171° 45' 40.09323" W	39.917m	417740.777	8471329.818	2.776m
PEL 2	0.001	0.004	13° 49' 35.73353" S	171° 45' 40.89441" W	40.049m	417716.752	8471320.892	2.905m

Table 4: Final AUSPOS Co-ordinates of All Stations Observed by Pelydryn.

3.4 Positioning Checks

3.4.1 Lever Arms

Following installation of the system into the aircraft, the Primary GNSS Antenna to IMU Reference Position (on top of IMU) Offsets are measured and input into the POSAV Controller software. On completion of the calibration flight the GPS lever arms are recalculated during post processing of the navigation data. The refined values are input into the controller software and checked each day on start up by the system operator. Following input of the post processed values no changes were observed during the course of the survey.

	Reference to IMU Lever Arms (m)	Reference to Primary GNSS Lever Arms (m)
X	-0.080	2.353
Y	-0.015	-0.297
Z	-0.126	-1.423

Table 5: Lever Arm Values

On a daily basis as part of the Quality Control on the navigation data the Lever Arms were checked to see if there was any drift in their values which would indicate movement of the system. **Figure 5** illustrates the values during the first sortie; a straight line in all 3 values shows there has been no changes in the lever arms used indicating that the converged values for the offsets are correct.

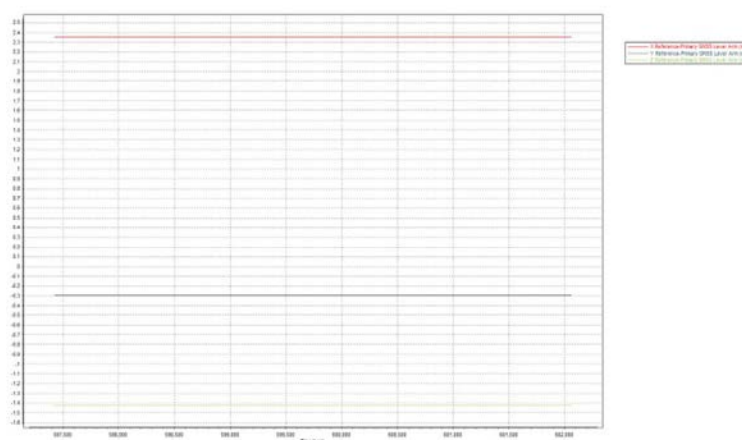


Figure 5: Lever Arms Checks in POSPAC MMS

3.4.2 Static Navigation Check

A Static Position Check was carried out on completion of each day's flying. The POSAV System in the HawkEye IIb is set by the operator to record data for 5 minutes after each flight whilst the aircraft is stationary. This data was analysed daily after processing and used to check that the POS AV is performing as expected with no noticeable instability in the observed horizontal and vertical positioning. The variation in the Northing, Easting and Ellipsoidal Height data at the end of each mission is assessed and used to give a gross error check. A sample output is in **Figure 6**, along with a sample comparison in the positions at **Table 6**.

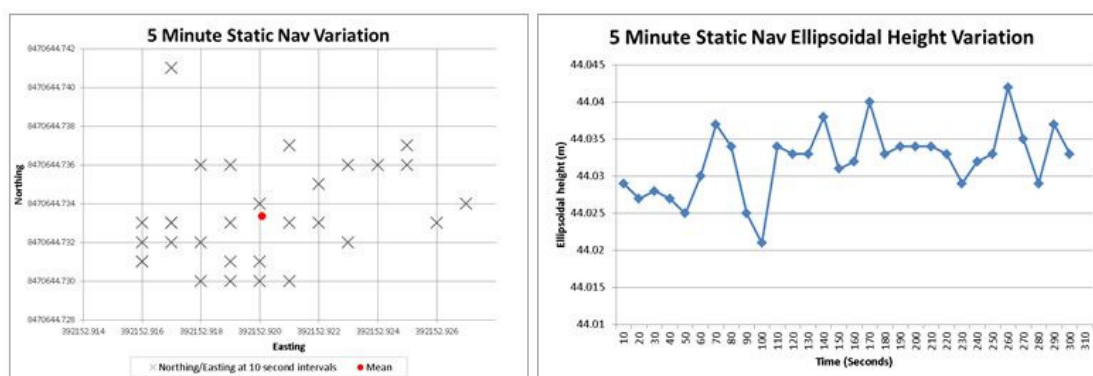


Figure 6: Typical Static Navigation Check Results

Statistical Summary			
	Easting	Northing	Ellipsoidal Height
Mean	392152.92	8470644.73	44.03
Standard Deviation	0.003	0.003	0.005
Min Value	392152.916	8470644.730	44.021
Max Value	392152.927	8470644.741	44.042
No. of Samples	30.00	30.00	30.00

Table 6: Example Results from Analysis of Static Validation.

Full details of the daily Static Navigation Checks for this survey are provided in Annex E of this report.

3.4.3 System Repeatability Check

3.4.3.1 Apia

A single area in the approaches to Apia Harbour was used as a local Sea Control Area (SCA) during survey operations (see Figure 7). As required by normal operating procedures, the Sea Control Area was overflown at the start and end of each flight to check the calibration of the system and ensure that there was no system 'drift' in the calibration values. Data from the calibration flight was used to compile a high resolution surface from the lines flown. Data from each subsequent sortie was processed and checked each day against this surface to ensure that the system was providing consistent results.



An aerial photograph of a forest with a red crosshair. The number 4178472 is displayed in red in the upper right quadrant. A small white 'x' is visible in the upper left quadrant. The forest is dense with green trees, and a road or path is visible in the lower left quadrant.

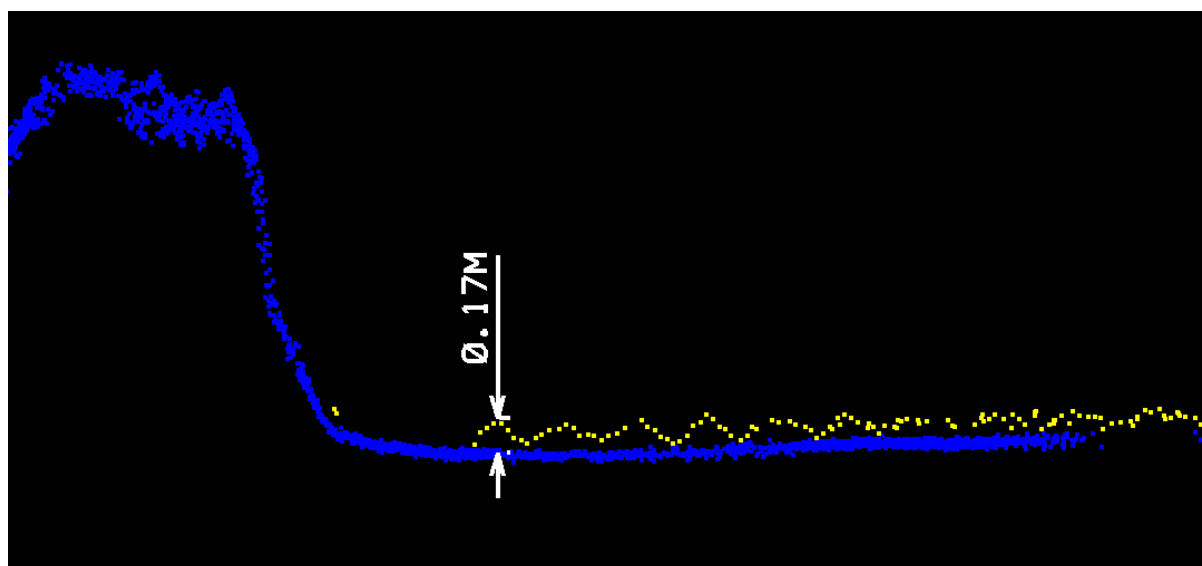


Figure 8: Apia Sea Control Area Overlain with LiDAR Data

Examination of the differences revealed a difference of 0.17m with a Standard Deviation of 0.71m. Lying within the stated requirements for IHO Order 1 the results were accepted and it was assumed that no changes in alignment had occurred during the transit from Vanuatu to Samoa. Daily Checks were then carried out to ensure no changes occurred during the survey.

Examination of the selected area allows the user to see if any drift is developing in the system's calibrations values. Due to the varying conditions experienced with water visibility, the results may vary slightly, but the check provides a continual check on errors and performance.

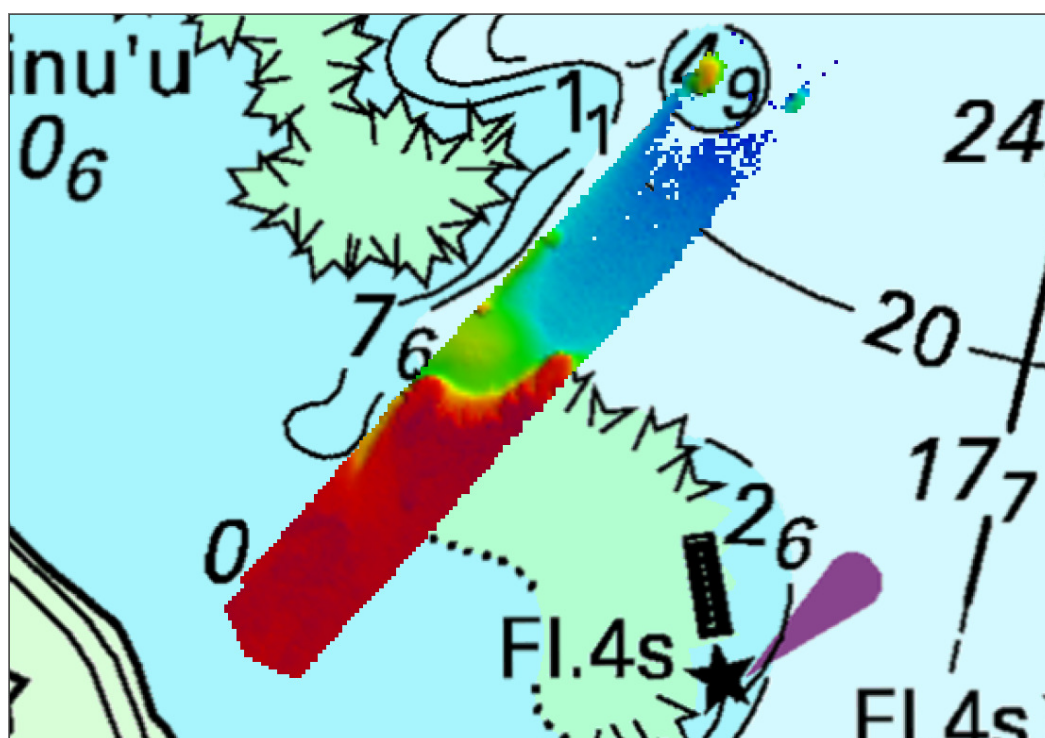


Figure 9: Dynamic Navigation Check Area

The following lines were flown during the course of the survey over this patch:

20121203_183344_Flightline 0_HD_exported.
20121203_213607_Flightline 0_HD_exported.
20121203_233746_Flightline 0_HD_exported.
20121205_184656_Flightline 0_HD_exported.
20121205_224944_Flightline 0_HD_exported.
20121207_180506_Flightline 0_HD_exported.
20121207_215748_Flightline 0_HD_exported.
20121208_003830_Flightline 0_HD_exported.
20121209_180654_Flightline 0_HD_exported.
20121209_202837_Flightline 0_HD_exported.
20121210_172750_Flightline 0_HD_exported.

Table 7: List of Lines flown over Efate Sea Control Area.

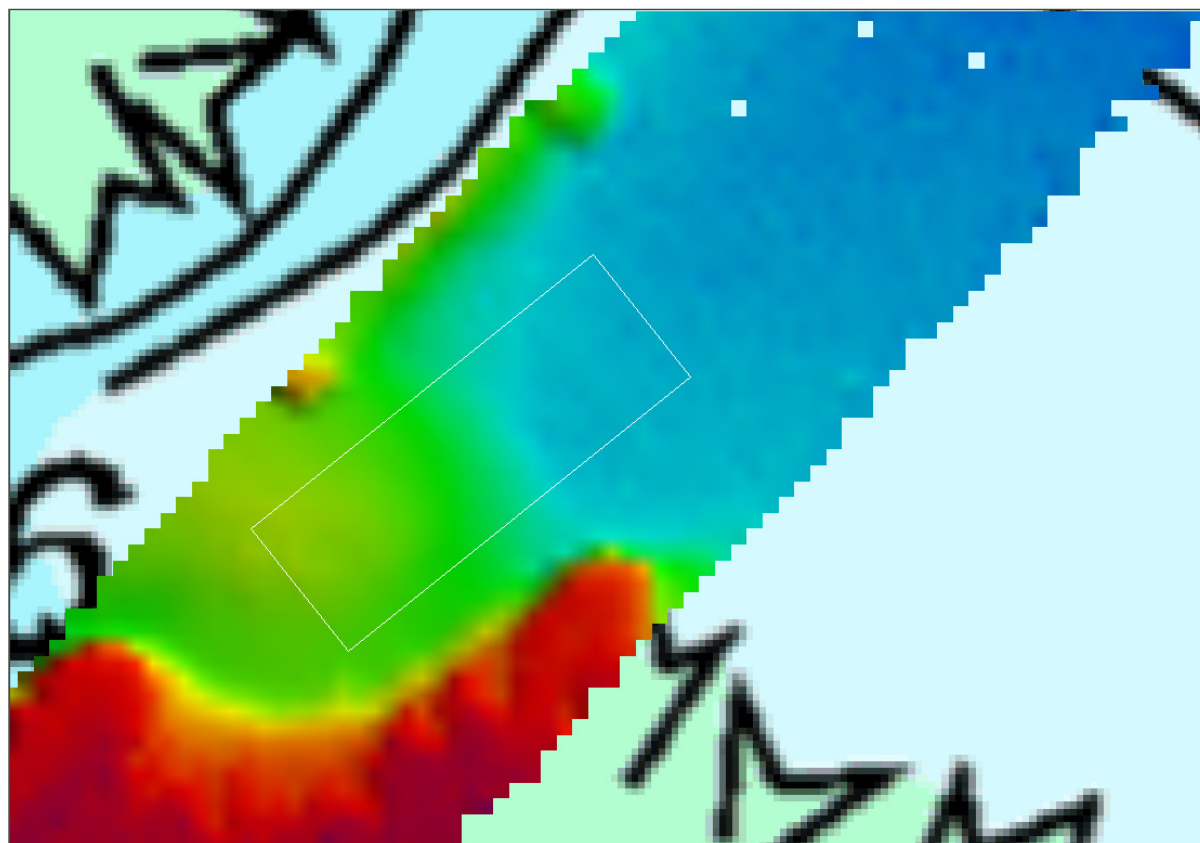


Figure 10: Close up of Apia Verification Area.

Figure 11 illustrates a cross section of the surfaces used for the dynamic checks

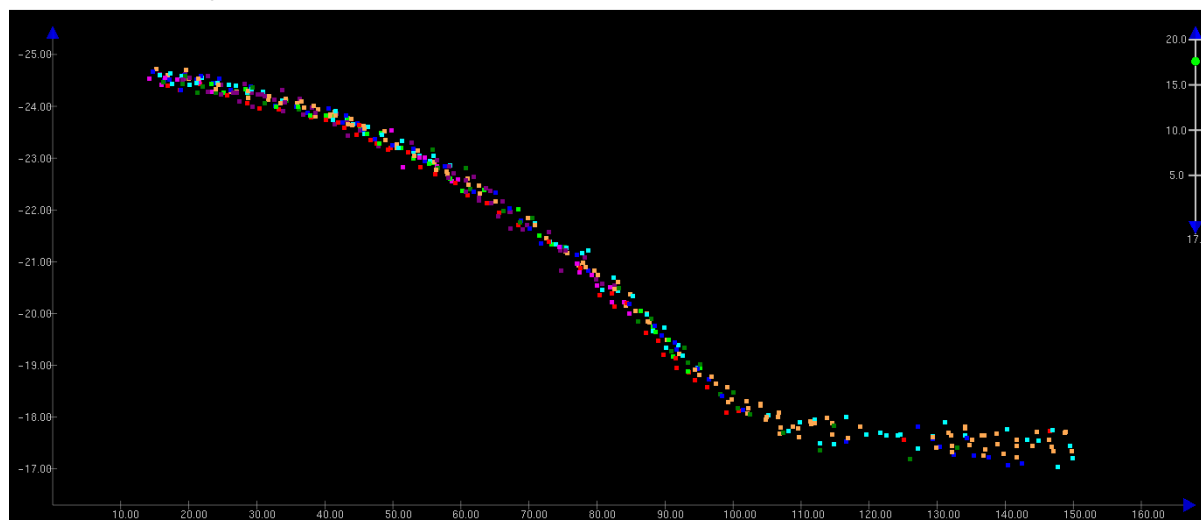
Repeatability Check Location**Long/Lat: 13.8205976 /171.7696488****Number of returns: 23904****Dimensions: 140/52m****Figure 11: Area 1 Cross Section Apia Update Image**

Figure 12 and **Table 8** show the statistical analyses of all lines subsequently flown over the Dynamic Check Area.

Area 1	Average dz	Std Deviation
Date	Metres	
20121203_183344	0.173	0.131
20121203_213607	0.084	0.126
20121203_233746	0.099	0.12
20121205_184656	0.111	0.135
20121205_224944	0.126	0.145
20121207_180506	0.268	0.141
20121207_215748	0.286	0.141
20121208_003830	0.162	0.128
20121209_180654	0.278	0.16
20121209_202837	0.118	0.125
20121210_172750	0.173	0.131
Average	0.17	0.13

Table 8: Summary of Repeatability Checks Apia, Samoa

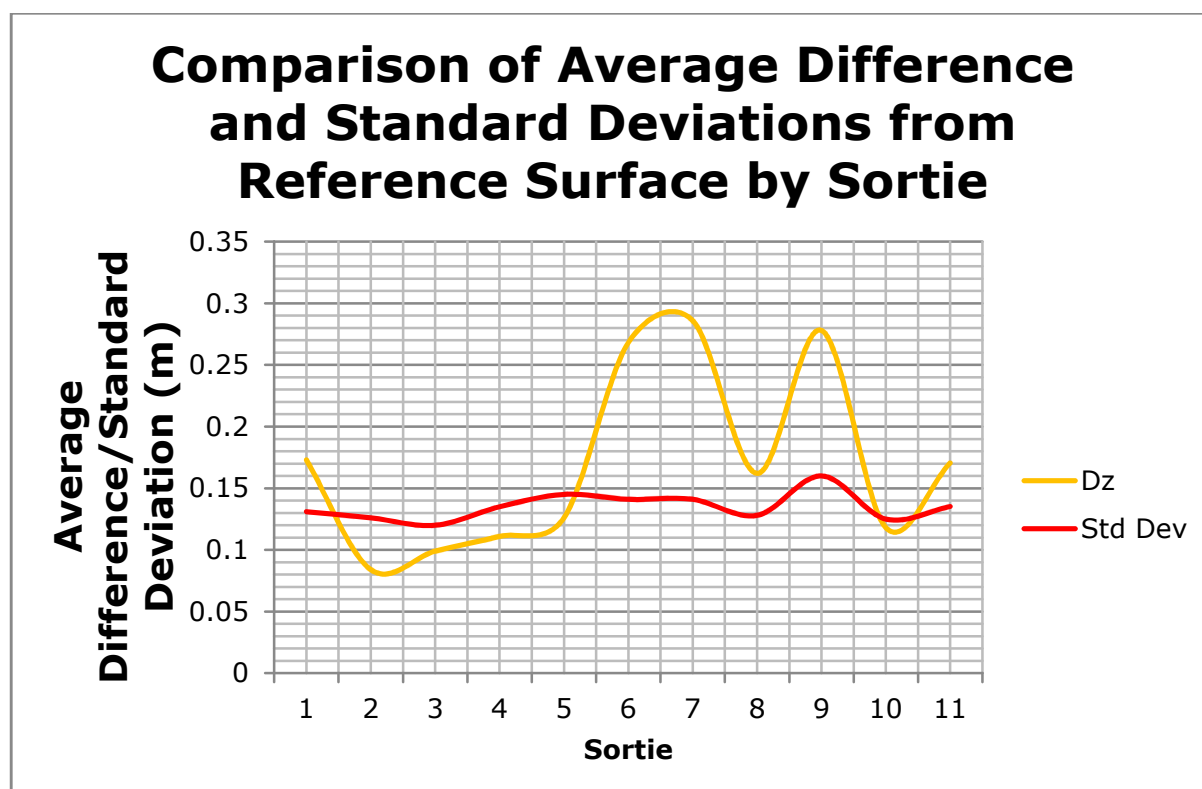


Figure 12: Statistical Results of Apia Dynamic Repeatability Check.

3.5 Vertical Control and Tidal Data

The data gathering phase of the survey was conducted on the ellipsoid in order to eliminate transformation errors in positioning and LiDAR acquisition. A single Tidal station was established during the course of the survey at the main port in Apia. A comprehensive tidal analysis is included in the Spatial Accuracy Report at **Error! Reference source not found..**

4 Data Acquisition Survey Operations

4.1 Introduction

An airborne LiDAR survey was undertaken in two areas along the coastal fringes of the Northern side of the Island of Upolu, Samoa between Mulifanua Wharf in the West and Vailele Bay in the East (**Figure 2**) to provide bathymetry up to the drying line and to provide topographic LIDAR data extending inshore to the limits defined within the Statement of Requirement.

Line spacing was set at 90m with general orientation running parallel to the coast. Crosslines were run across each block to allow checking of the application of the ellipsoidal height solution.

4.2 System

The survey was undertaken using the HawkEye IIb bathymetric LiDAR system. System 002 was used throughout the survey. One minor defect impacted on operations causing some delay to the programme:

- a. System Air Conditioning Unit Failures due to loss of gas and controller board failure.

As a result of this, the system tended to overheat late in each sortie and careful monitoring and flight management had to be introduced in order to keep the system in its optimum operating temperature range.

The survey was flown on Flight Case 250. Flight plans were optimised to allow the longest lines possible to reduce the number of turns required. All reflights were conducted on Flight Case 250. Full details of flight plans and flight cases can be found at Section 4.3.1.



Figure 13: HawkEye System 002 installed in NOMAD

The HawkEye IIb was calibrated to the aircraft by means of a topographic Ground Calibration Area

and a Sea Calibration Area, both in Port Vila. Details of the Calibration are found in the Mobilisation Report. The calibration and validation results indicate that the HawkEye IIb system was operating within specified parameters and achieved the required accuracy for the survey.

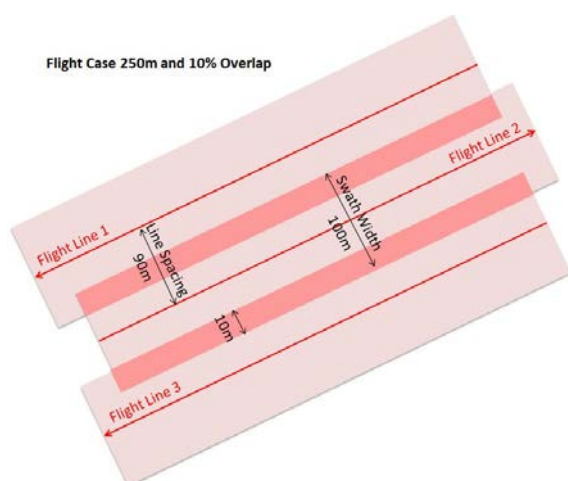
4.3 Survey Parameters

LiDAR used	:	HawkEye IIb
Operating frequencies (kHz)	:	1Khz (Bathymetric) 4Khz (topographic)
Horizontal datum	:	ITRF 2008
Spheroid	:	WGS 84 (GRS80)
Projection	:	UTM (S)
Central Meridian and Grid Zone	:	Zone 2
Vertical datum	:	Survey on Ellipsoid
Main Line Spacing (m)	:	90m (FC 250)
Line Direction	:	Varies (Parallel to Coast)
Cross Line Spacing	:	1700m
Cross Line Direction	:	Perpendicular to Coast
Average survey speed (knts)	:	130knts
Average across track % overlap	:	10%
General water depth achieved (m)	:	35-40

Table 9: Survey Parameters - Samoa

4.3.1 Flight Case Parameters

The HawkEye IIB has several flight cases. To achieve 100% coverage, and ensure that the required standards for the survey were achieved, the flight case 250 was used (**Figure 14**). Main lines for the LiDAR data acquisition were conducted at a line spacing of 90m; the flight altitude was set at 250m with a swath width of 100m.



Flight Attribute	401 Scan Case
Flight Case Name	250
Altitude (m)	250
Speed (m/s)	67-74
Swath Width (m)	100
Lateral Shot Distance (m)	2.9
Forward Shot distance (m)	2.9
Percentage overlap (%)	10
Line spacing using above overlap (m)	90

Figure 14: Flightcase 250 Details.

4.3.2 Apia Area B Survey Configuration

Apia Area B Survey Configuration

Mission Name: Apia_B			
Horizontal Datum:	WGS84	Vertical Datum:	Ellipsoidal
Flight case:	250	Altitude:	250m
Swath Width:	100m	Line Spacing:	90m
Speed:	110knts	Coverage:	110%
Number of Lines:	54	Total Line Length:	373km

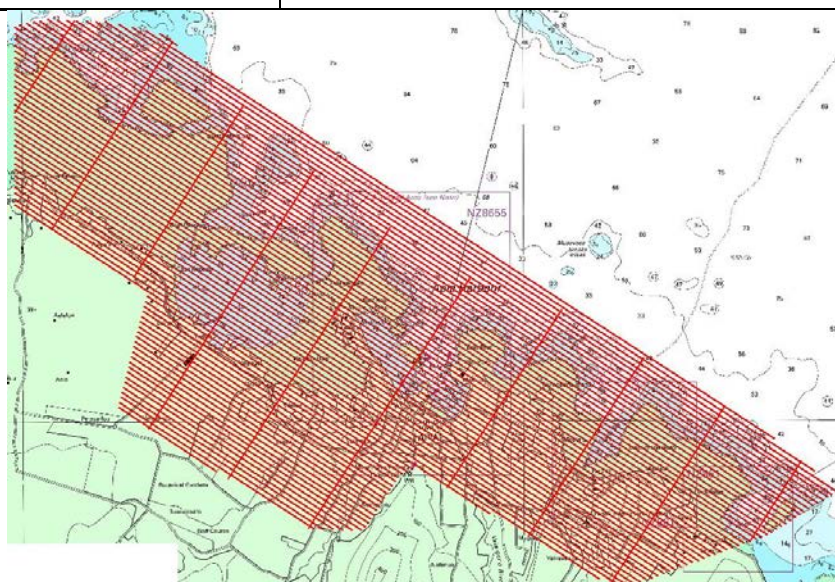


Figure 15: Apia Area B Line Planning Information.

4.3.3 Apia Area A Survey Configuration

Apia Area A Survey Configuration

Mission Name: Apia_A			
Horizontal Datum:	WGS84	Vertical Datum:	Ellipsoidal
Flight case:	250	Altitude:	250m
Swath Width:	100m	Line Spacing:	90m
Speed:	110knts	Coverage:	110%
Number of Lines:	78	Total Line Length:	827.5km

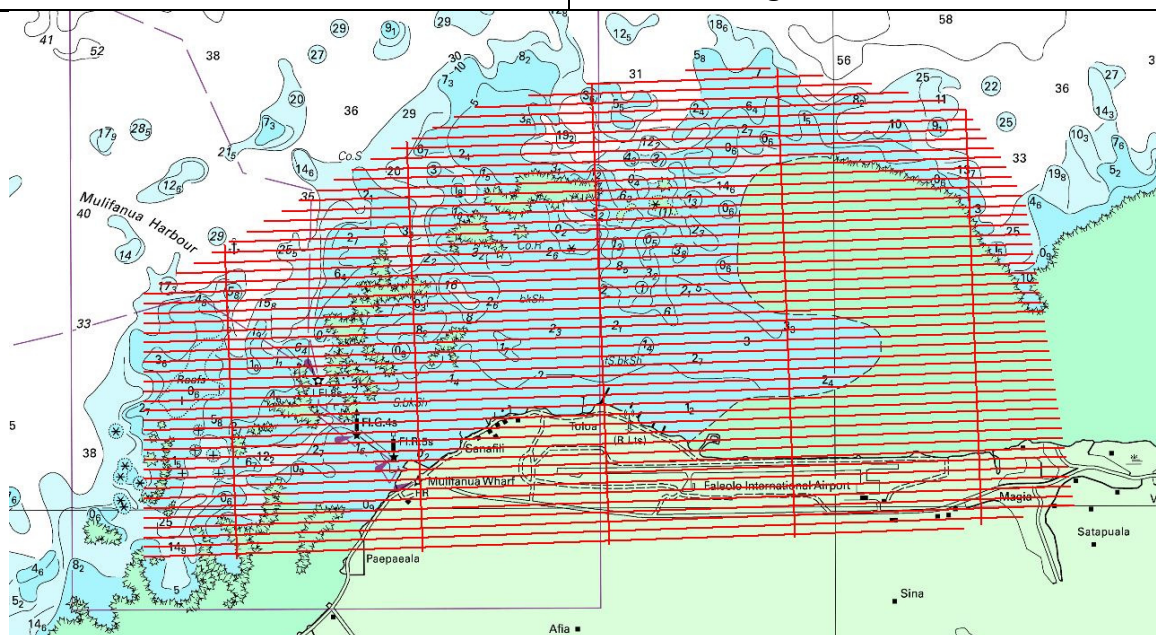


Figure 16: Apia Area A Line Planning Information.

Upon completion of the planned first pass lines, data coverage was analysed in the field and reflights were flown to improve coverage where the turbidity, sun glint and white water affected light penetration. Some lines were reflown to improve coverage where the aircraft was off track due to turbulence.

4.4 Data Logging Parameters

The flying altitude and flight case establishes the scanner logging parameters and is set during mission planning. Although it is possible to modify it during a flight for each individual flightline, no amendments were made on this project during flight.

4.4.1 Hydro laser

Parameters are set and modified during a flight in the shallow and deep receivers in order to maximise the efficiency of the gain (or monitor signal).

The following parameters can be altered: K, Go and Range Time Correction (see **Table 10**). The K

parameter represents the diffuse attenuation coefficient which increases as water clarity decreases. The gain (Go) is used to compensate the damping of light in water.

The laser return data is recorded during a time window (in nanoseconds) that is manually set. The part of the waveform saved is the start of the laser pulse and the response from water surface and bottom. The time between the start of the pulse (TO) and return from the surface is the Range Time Correction.

Area	K shallow	K deep	Go shallow	Go deep
Apia A	0.8 - 0.17	0.7 - 0.15	-2.5 - -1.7	-2.5 - -1.6
Apia B	0.17 - 0.17	0.11 - 0.15	-2.9 - -2.1	-2 - -1.6

Table 10: Hydro Laser Minimum and Maximum Parameters

4.4.2 Topo Laser

The operator can set and monitor the Topo laser power during acquisition. Power is generally set to 2500mW during planning. The power can be changed during flight for each individual flightline. Topo power was reduced over highly reflective areas during this project.

4.4.3 Camera Parameters

The operator can set and monitor the exposure time during acquisition. This parameter varied from 0.5-2 on bright days to 3-5 on normal days.

4.5 Software

The following software was used during the course of the survey:

System Acquisition Software:

- AOC Version: 1.07.03

Processing Software:

- CSS: Version 2.03.11
- PosPAC MMS Version: 5.4 SP 1
- CARIS HIPS and SIPS Version: 7.1 Service Pack 2 Hotfix 6
- Fledermaus Version 7.3.2a
- TerraScan Version 12

4.6 Environmental Impacts on Survey Operations

4.6.1 Meteorological Impacts

Weather had a significant impact on operations due to the high rainfall experienced, with the accompanying low cloud and wind squalls, which prevented or curtailed flight sorties on a regular basis. A number of short re-flights were required due to the aircraft being off track as a result of strong winds and turbulence in the survey area.

On a number of occasions low cloud (below 1200ft) meant surveying operations had to be suspended or the flight sortie moved due to the scattering effect of water droplets in the cloud.



Figure 17: Low Cloud along the coastline.

4.6.2 Sun Glint

Sun glint is a known issue with bathymetric LiDAR and line direction is, wherever possible planned to reduce the impact. Sun glint increases the amount of noise in the data which becomes more significant when weak bottom returns from deeper areas combine with areas of high turbidity. It can also impact in shallow water on areas of light seabed. Sun glint was therefore a significant factor when surveying the deeper areas later in the period when the sun angle remained higher than 60 degrees for longer periods during the day. Operations were conducted in the early and late parts of the day at the inception of operations, but as weather conditions impacted on the survey it was necessary to fly between rain showers and this has impacted on data quality on occasions.



Figure 18: Sun Glint along the shallow coastal fringe.

4.6.3 Extreme Shallow Water

Designed to operate across the land/sea interface, the system generally can detect depths up to 0.5m. In most areas the range of tide allows either a deeper return at a higher state of tide or a topographic detection at low water. The Tidal range experienced in Samoa was 0.7-1.2m during the survey, which meant that the shallow fringing reef plates would be difficult to survey if there was any significant wave action. A number of factors contribute to the detection of a valid return with the laser in extreme shallow water situations:

- a. Waveform Discrimination - The shallow water algorithm does not work well in these situations and detection will drift in and out and can cause variation in the depths seen due to the surface and seabed return being so close together on the waveform and the system's ability to discriminate between the two.

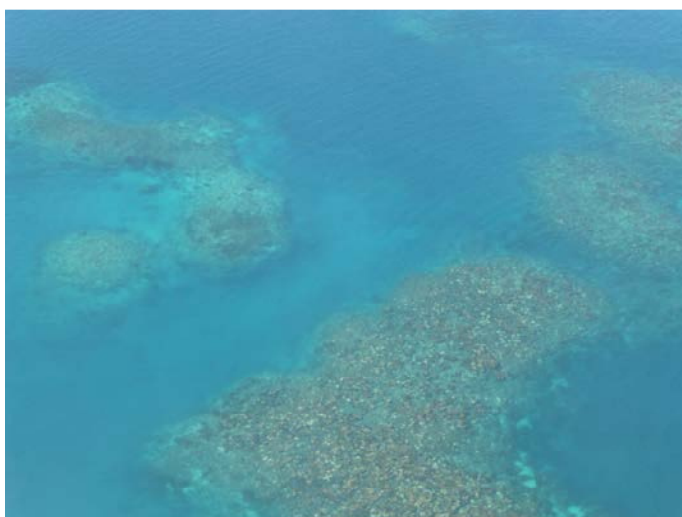


Figure 19: Extreme Shallow Water over Coral Reef

- b. Slope - Steep changes of slope when we go from shallow to deep along the reef drop offs means the system's operating parameters are changing rapidly between the shallow and

- deep water algorithms, where there are large slopes this can impact on detection
- c. Very Shallow water can cause extreme glint even at low altitude (**Figure 200**), the resultant reflections can reduce the quality of the surface and seabed return.
- d. Change in Classification – i.e. when going from Land to Sea next to jetties and breakwaters (**Figure 20**). The often rapid changes in between land and water as you pass over small islands or manmade structures can lead to a lag in the system’s ability to track the seabed temporarily on the leeward side of the feature.



Figure 20: Extreme Shallow Water with topographic features cause problems in seabed tracking and the effects of Sun Glint in shallow water.

- e. Seabed Contrast. Very dark patches or black areas such as weed or rock or light areas such as dead coral heads along with a surrounding highly contrasting seabed, (very shallow to deep water) can impact on the system’s ability to classify the return resulting in data gaps in areas where depths may be expected (**Figure 211**). Changes in the colour of the seawater can also impact on the quality of the return.



Figure 21: Contrast and slope changes behind or close to features affect seabed detection.

As a result of these known limitations, lines were flown in opposite directions and at different times of day and states of tide to mitigate most of the effects noted above. Following field cleaning and analysis of the acquired data a number of gaps were found on the coastal fringing reef plate. These areas were reflown in an attempt to infill the gaps seen and provide as complete a dataset as possible.

4.6.4 Turbidity

Turbidity severely impacts on a bathymetric LiDAR’s ability to achieve penetration in any depth of

water. Typically achieving seabed detections at 2.5 times Secchi disk depth, turbidity can reduce this significantly. Heavy rainfall or coastal works can cause significant sediment run off into local waters. Tidally modified, these sediment plumes can drift up and down the coast reducing the ability to detect the seabed. On both islands this became a significant problem and attempts were made to mitigate the effects by flying at different states of tide or flying additional lines. The water to the west was found to be significantly more turbid despite the lack of rivers to contribute to run-off. This was inferred to be a result of plankton blooms which were moving with the tides and therefore causing variations in the penetration achieved.



Figure 22: Examples of Turbidity caused by Surface Water Run-off.

5 Data Processing

5.1 Field Processing

Pelydryn undertook data cleaning whilst in the field to quality control data and assess density and coverage for reflight purposes. In order to do this it was necessary to process the raw waveforms with post processed navigation to produce LAS files. These files were imported into CARIS, following QC in Coastal Survey Studio, where they were checked for density, coverage and vertical misalignments. Topographic data along with imagery were also checked to allow any flight replanning to be undertaken.

5.1.1 Data Backup and Archiving

On completion of each day's flying, raw navigation and HEII data was saved to a Back-up disc and a Processing disc. Any subsequent field processing of data was carried out only on the data on the Processing disc. The following data was nominally backed up on Portable USB Hard Drives

- a. Raw HawkEye Wave Form Data and Imagery.
- b. POSAV Trajectory Files.
- c. GPS Base Station data.
- d. Processed Waveform Data (following merge with post processed navigation data).
- e. Smoothed Best Estimate of Trajectory Files (SBET) for use with imagery processing.
- f. Time referenced imagery for image mosaicking.
- g. LAS Files for Bathymetry, Topography and not Found data outputs from Coastal Survey Studio.
- h. Processing Reports and Forms.

5.1.2 Data Processing

Raw aircraft navigation files were processed using PosPAC MMS to produce a post processed navigation solution utilising rapid ephemeris and a ground based GPS Station data which lay within 30km of the area being surveyed. Figure **23** below illustrates the typical workflow for data processing used on this project. Following navigation processing the post processed inertial navigation solution is merged with the raw waveforms to produce a processed waveform in CSS. This waveform is then exported as a LAS File for subsequent data editing in CARIS HIPs. Concurrently the imagery taken during survey is time tagged and renamed to allow an ortho-mosaic to be produced at a later stage.

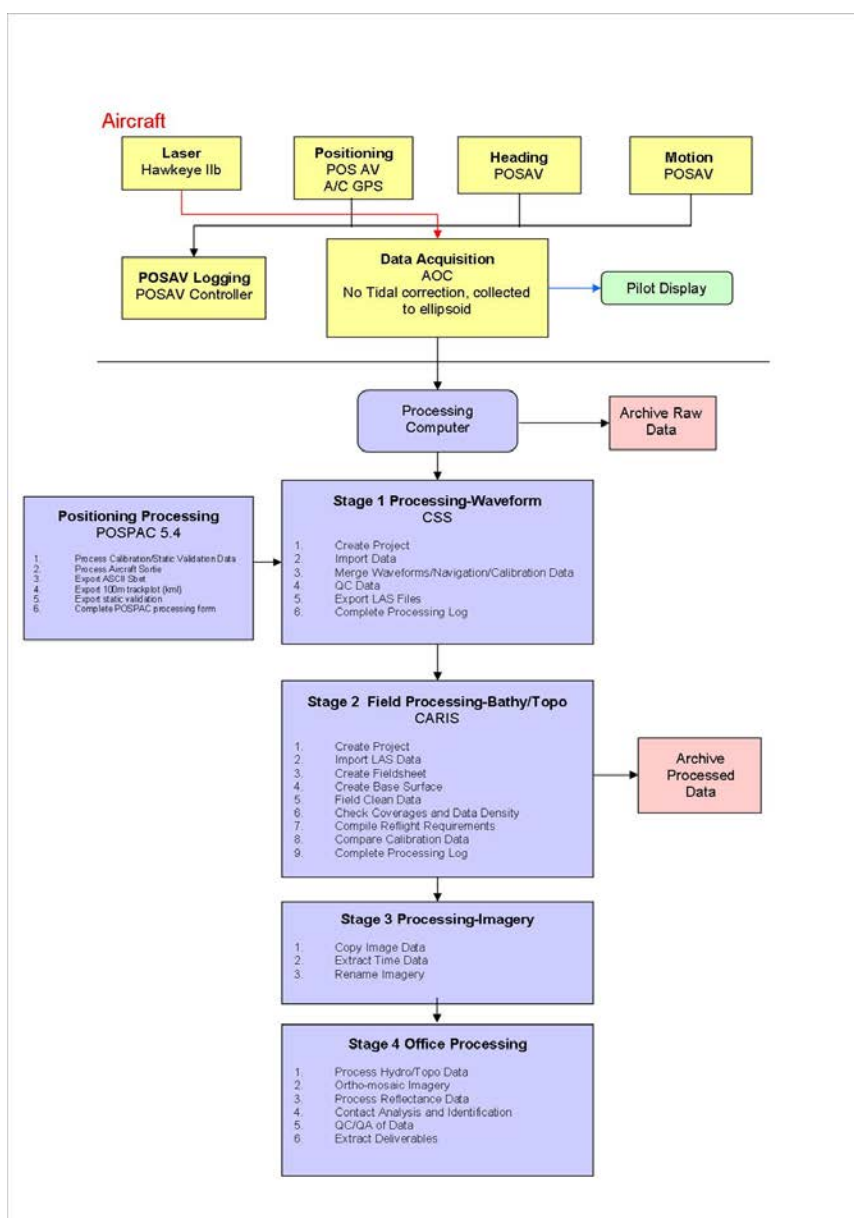


Figure 23: Processing Workflow for Samoa

5.1.2.1 Navigation Post-Processing

The PosPAC MMS software applies data from the installed Applanix Inertial Measurement Unit (POS AV 410) to produce a Real Time Vehicle Frame (RTVF) navigation solution. Base Station observables (from a co-ordinated station) in the form of RINEX data along with ephemeris data is imported into PosPAC MMS to give a decimetre accuracy solution to the GPS position. PosPAC MMS creates a Smoothed Best Estimate Trajectory (SBET), which combines the Corrected GPS position and the IMU data to produce a navigation solution with centimetric accuracy.

The navigation solution is exported from PosPAC MMS in text file format and used as an input to processing the raw HawkEye IIb files in CSS. It is also exported as an SBET for imagery processing.

5.1.2.2 Waveform Processing

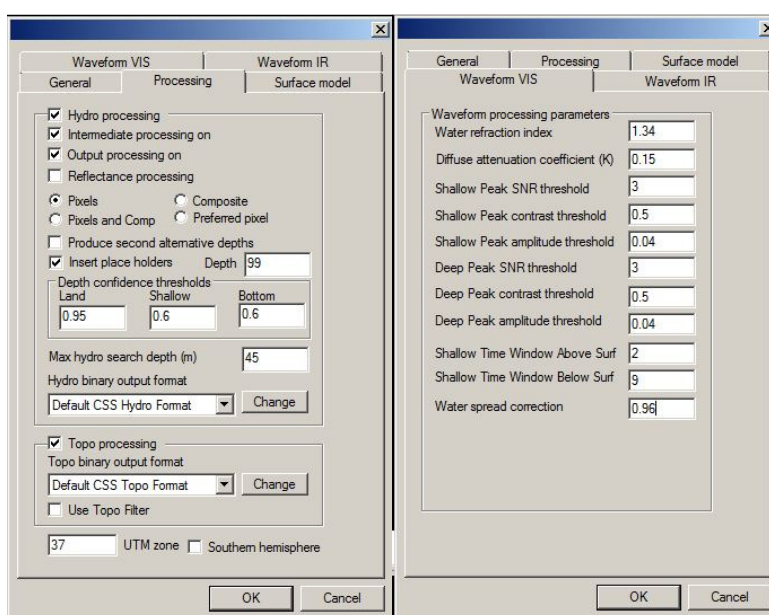
Coastal Survey Studio (CSS) is the propriety processing software for processing HawkEye IIb data. It

takes inputs in the form of a navigation file, a calibration file and the raw wave forms from the HawkEye IIb. The output from CSS is *.LAS files which can be imported to other processing and visualisation software for subsequent data editing.

CSS processes the wave forms to produce values classified by laser into 'Deep', 'Shallow', 'Land' or 'No Bottom'. These values can be output as Ellipsoidal Heights or depths. Outputs can be set by the processor but currently all data variables are output into the CSS files.

Batch Processing of HawkEye IIb files requires the input of 'confidence' settings and 'k' values by the Processor. These are initially set based on conditions experienced during the survey. Inspection of individual waveforms, especially those which return 'No Bottom' values, is carried out after batch processing to check the automatic interpretation of wave forms. The 'confidence' setting can be adjusted where wave forms show evidence of returns that have not been classified by the automatic processing. The effect of this can be to introduce more 'noise' into the data but also more valid 'Bottom' returns. The Processor will adjust the 'confidence' value and reprocess when it is considered advantageous to the overall survey.

A summary of typical processing parameters is in Figure 24.



The figure displays two side-by-side screenshots of the CSS software interface, showing various processing parameters.

Left Window (Waveform VIS / Waveform IR):

- General Tab:**
 - ☒ Hydro processing
 - ☒ Intermediate processing on
 - ☒ Output processing on
 - ☐ Reflectance processing
 - ☒ Pixels ☐ Composite
 - ☐ Pixels and Comp ☐ Preferred pixel
 - ☐ Produce second alternative depths
 - ☒ Insert place holders Depth: 99
 - Depth confidence thresholds:

Land	Shallow	Bottom
0.95	0.6	0.6
 - Max hydro search depth (m): 45
 - Hydro binary output format: Default CSS Hydro Format
 - ☒ Topo processing
 - Topo binary output format: Default CSS Topo Format
 - ☐ Use Topo Filter
 - UTM zone: 37 Southern hemisphere
- Processing Tab:** (Empty)
- Surface model Tab:** (Empty)

Right Window (General / Processing / Surface model):

- General Tab:** (Empty)
- Processing Tab:**
 - Waveform processing parameters
 - Water refraction index: 1.34
 - Diffuse attenuation coefficient (K): 0.15
 - Shallow Peak SNR threshold: 3
 - Shallow Peak contrast threshold: 0.5
 - Shallow Peak amplitude threshold: 0.04
 - Deep Peak SNR threshold: 3
 - Deep Peak contrast threshold: 0.5
 - Deep Peak amplitude threshold: 0.04
 - Shallow Time Window Above Surf: 2
 - Shallow Time Window Below Surf: 9
 - Water spread correction: 0.96
- Surface model Tab:** (Empty)

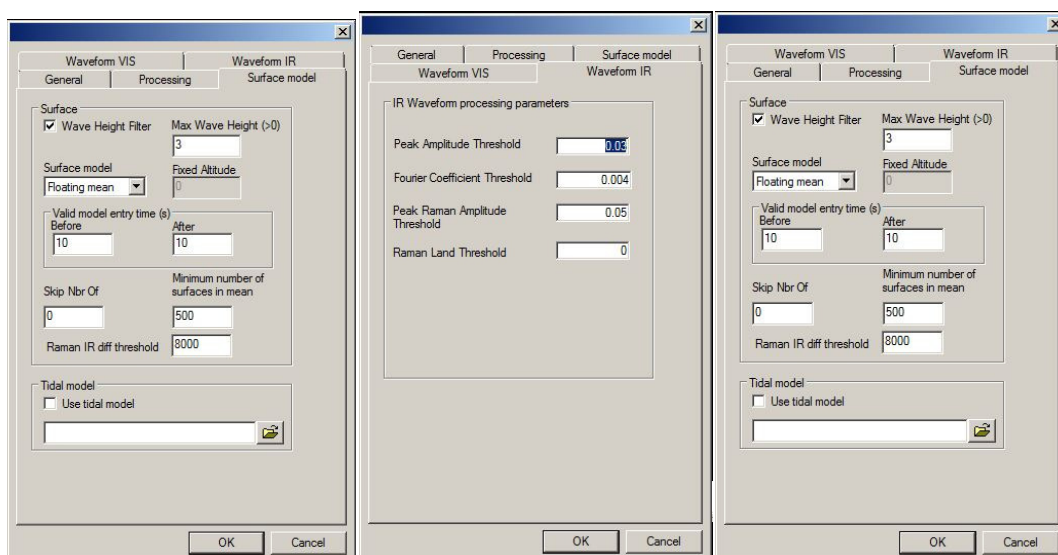


Figure 24: Typical CSS Processing Settings

As part of Pelydryn's Quality Control, each line processed in CSS is visualised to check for coverage, beam width and scan pattern, as well as a gross error check. Once lines are accepted they are exported as LAS files. It is possible to filter the 'no bottom' values at this stage and they can be removed from the processed data set.

LAS files were exported from CSS in the form of Point Latitude, Point Longitude and Height. Three LAS file types were output:

- LAS_HD – Bathymetric Data
- LAS_NotFound – All data classified as not valid
- LAS_TD –Topographic Data.

5.1.2.3 Field Cleaning and Quality Control of Hydro Data

All LAS HD (Hydro) files were imported into CARIS HIPS (Version 7 Service Pack 2 Hotfix 6) to check coverage, data quality and navigation. Pelydryn undertook basic filtering and cleaning to remove outliers, most notably those in the form of 'surface returns' detected by LiDAR. The following procedure was used to verify the data collected:

- LAS Files were imported into CARIS HIPS on a day and area basis.
- The sortie to be processed was loaded with a chart backdrop to remove surface noise and outliers.
- Once complete a 2m base surface was made and the sortie is checked for separation areas and outliers using the Standard Deviation Surface.
- Once accepted the data is combined with all other data for that area and a Base Surface for the area is created or updated at 3m and 30m resolution.
- The data is checked to ensure that there are no discrepancies between lines and any areas of poor coverage or density are noted for reflight.
- The Field Manager confirms the lines entered agree with the Flight Log, Processing Log and CARIS (This ensures all valid data is included).
- A check using the Chart is then carried out to identify any new shoals or the extents of

coverage achieved. The Density and coverage checks are reviewed and any lines to be reflown or areas for reflight are created.

5.1.2.4 Field Cleaning and Quality Control of Topo Data

Field cleaning of the topographic LiDAR Data was undertaken in the same way as the Hydro data stated in Section 5.1.2.3 above. Greater attention was paid to along track overlaps and alignment to ensure that no corrections were required between lines to ensure a seamless dataset.

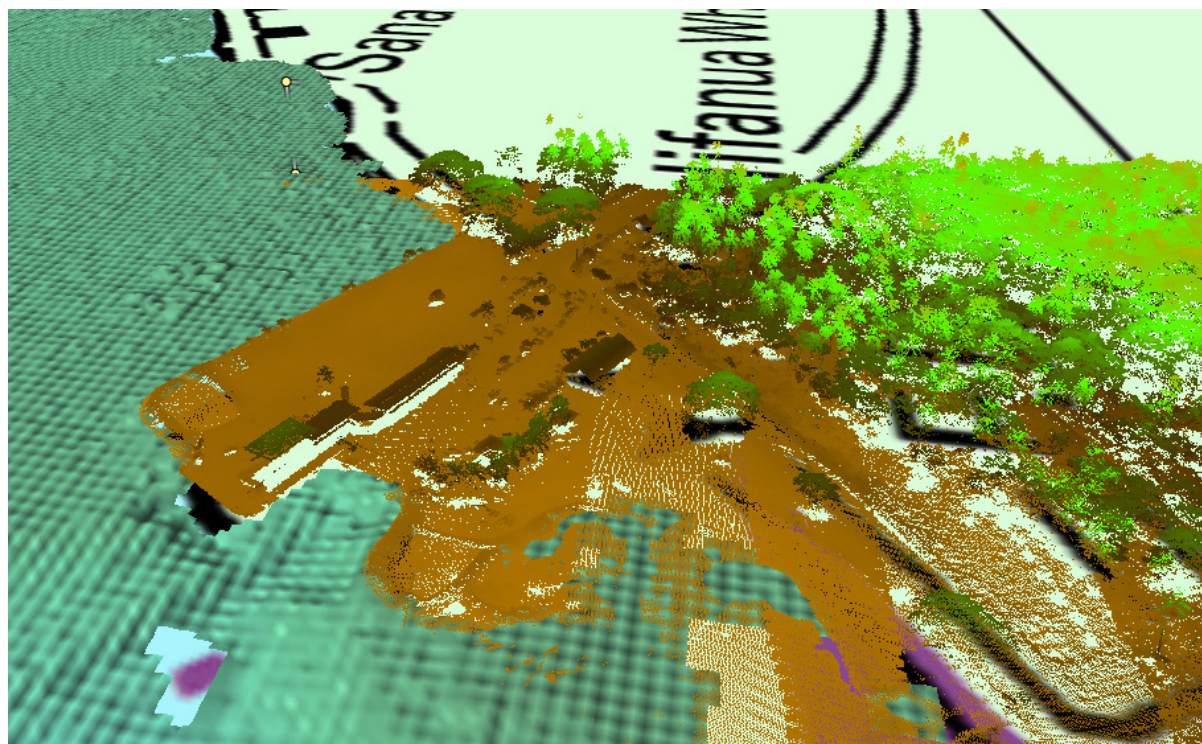


Figure 25: Point Cloud of Topographic LiDAR over Mulifanua Wharf viewed in Fledermaus.

5.2 Office Processing

Once the survey was complete the data was transferred to the United Kingdom for final data processing. Once backed up, data processing was completed using CARIS HIPS. The following datasets were transferred to AAM for processing and analysis:

- a. Unclassified topographic data in LAS file format.
- b. All imagery captured.
- c. All trajectory files for imagery processing.

5.2.1 Ortho Mosaic Processing

Ortho-mosaics were processed by Pelydryn utilising Terraphoto and have been supplied as a deliverable.

5.2.2 CARIS

The raw LAS files were previously imported in to CARIS HIPS using the Conversion Wizard and field cleaned. On arrival in the office, new Base Surfaces were calculated and the swath data was

examined with noise and outliers rejected. Data Processing was conducted in three stages:

- a. Process Bathymetric Data up to the waterline.
- b. Process Topographic Data down to the waterline.
- c. Process the Coastal overlap using both the Bathymetric and Topographic Data.

Data was processed at a higher resolution than the final deliverable in order to ensure that all outliers were identified and flagged rejected. Base Surfaces at 3m resolution were built with a full set of statistics allowing the generation of Shoal Deep and Standard Deviation Surfaces, which could be used to identify data for removal. Data was processed in 1km blocks using the three surfaces highlighted above to identify data for rejection. In the coastal overlap, imagery taken at the time of flight was also used to identify features and the location of the breaker line to assist in the removal of data in these areas. All contacts and significant features identified in processing were designated and recorded in the Contact Log.

5.2.3 Bathymetric Reflectance Processing

Reflectance data was a required deliverable for this project and peak amplitude data was extracted from the raw waveforms to allow data processing to take place following the following steps:

1. Following processing of the raw waveforms in CSS the Peak Amplitude of each point was exported in LAS Format in xyza format
2. The exported LAS files were imported into Terrascan
3. The resultant dataset intensity values are adjusted to allow for different incident angles and differences in return strength due to gain and altitude variations
4. The resultant values were scaled for a 0-256 8 Bit color map and the final contrast adjusted.
5. The resultant data was exported as an ECW and in xyzi format

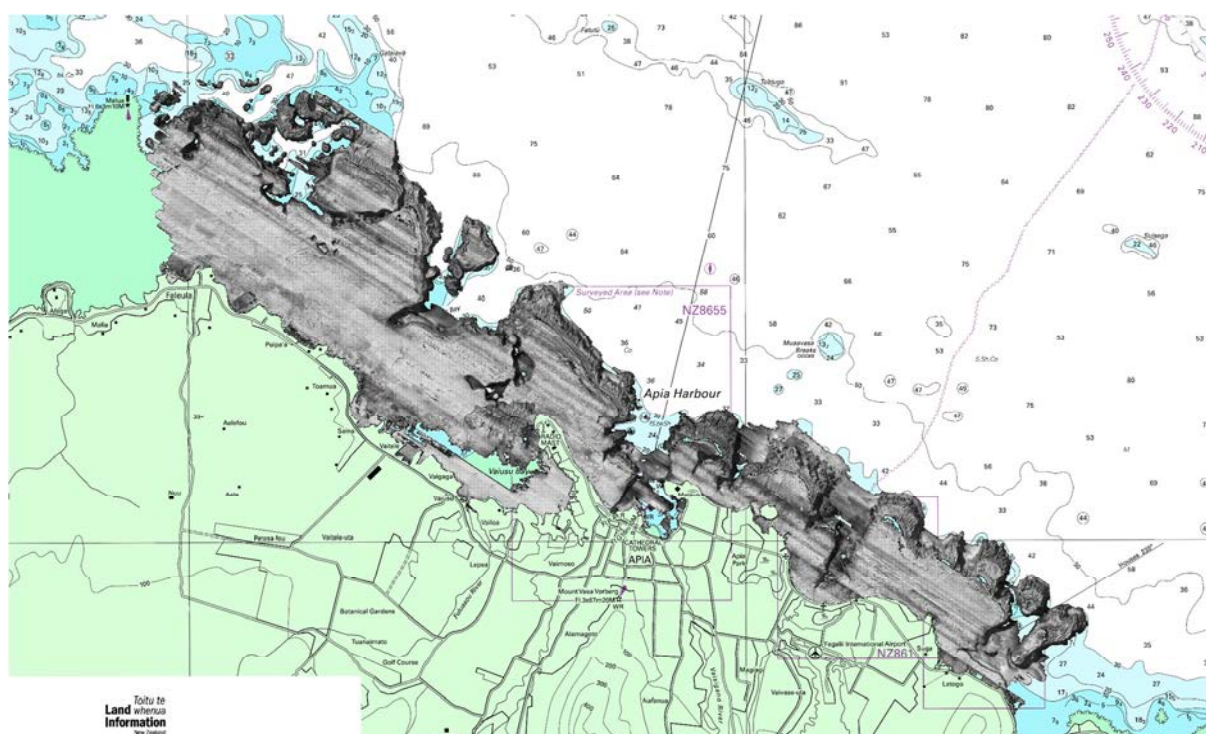


Figure 26: Sample Reflectance Data from Area B with underlain bathymetric chart.

Data is supplied with this report in both ECW and XYZi at 5m resolution.

5.2.4 Quality Control and Analysis of Data

Following data cleaning the data was quality controlled by a senior member of Pelydryn's staff in order to ensure no spurious data remained. All Quality Control and Analysis of the data was undertaken using Fledermaus. The CARIS HDCS datasets were imported into DMagic (Data preparation tool) and a PFM (3d Visualisation surface) was built. The resultant surfaces were loaded into Fledermaus where they were checked and if required edited to remove any outliers that had been missed during the processing phase. Once these checks were complete the data was unloaded back to CARIS and final base surfaces were built at the resolution of the final product to allow final checks of the data to be carried out.

5.2.5 Extraction of Data Deliverables

Following QC/QA of all point data, all accepted points were exported from CARIS in ASCII format. This data was then imported and merged with both the Reflectance Data and Classification to derive the two final datasets referenced to the ellipsoid.

Once the data was merged it was split into 1km blocks based on UTM Grid Zone 2S for Samoa using TerraSolid. As part of the QA process this data was reimported into Fledermaus and a surface comparison completed to ensure all data had been exported and merged correctly. Data in xyz format and Shapefiles defining the block accompany this report.

5.2.6 Digital Elevation Model (DEM)

ArcGIS Terrains were created for ground and seabed classified LiDAR data. One metre (ground) and five metre (seabed) ESRI ArcGIS binary grids were derived using a customised ArcGIS macro. A seamless DEM was created as 5m ESRI binary grids integrating the land and sea data.

The seamless DEM has been generated using the same technique as adopted for the Vanuatu dataset and accepted by Geoscience Australia.

5.2.7 Digital Surface Model (DSM)

ArcGIS surfaces were created from non-ground classified LiDAR data. One metre ESRI ArcGIS binary grids were derived using a customised ArcGIS macro.

5.2.8 Canopy Elevation Model (CEM)

A canopy elevation model was produced as two metre ESRI ArcGIS binary grids using ArcGIS with the following methodology:

- All ground surface derived using natural neighbours interpolation [G] (*Floating Point*)
- First return vegetation surface derived using natural neighbours interpolation [V1] (*Floating Point*)
- Height in metres above ground of first vegetation returns $[V1] - [G] = [H]$ (*Floating Point*)
- Mask of all vegetation ALS returns with 1m buffer, reclassified to 0 outside mask and NoData inside [M] (*Integer*)
- Mosaic of canopy height surface and vegetation mask $[H] + [M] = \text{CEM}$ (*Floating Point*)

5.2.9 Foliage Cover Model (FCM)

A foliage cover model was produced as 10 metre ESRI ArcGIS binary grids using ArcGIS with the following methodology:

- Count of all vegetation returns per 10m cell [V] (*Integer*)
- Count of total ALS returns per 10m cell [A] (*Integer*)
- Proportion of all vegetation to total ALS returns expressed as a percentage $([V] / [A]) * 100 = \text{FCM (Floating Point)}$

5.2.10 Contours

Contours were generated from both bathymetric and topographic data in ESRI shape file format with 0.5m and 5.0m intervals. Contours were also generated for the seamless dataset.

5.2.11 Topographic Intensity Imagery

Due to the fact that both topographic and bathymetric data was captured with the Hawkeye IIb bathymetric LiDAR system it was necessary to generate the topographic intensity imagery by a different technique to that adopted by the topographic LiDAR systems. The following technique was adopted.

1. Using terra photo, attach all the topographic imagery data together in "**manage raster references**" tool.
2. In TerraScan PRJ, change the directory folder into the copy of TP_Adjusted folder. Turn on the **colour** box in attributes before save the TerraScan PRJ.
3. Select **extract colour from images** from TerraScan PRJ Tools and run for all blocks.
4. Run macro to output CENHR- Class (all classes except cut overlap), Easting, Northing, Height, Red channel in xyz format from LAS data extracted.
5. Import CENHR dataset into tiles and generate the intensity images from the LAS data imported.

5.2.12 Project Plan

Prior to data capture commencement, a Project Plan was submitted to and approved by Geoscience Australia. This Project Plan details work breakdown structure, agreed data capture plans, project milestones and delivery schedules, progress reporting schedules.

5.2.13 Pre Survey Quality Assurance Plan

Prior to the commencement of aerial LiDAR data capture, AAM prepared and submitted to Geoscience Australia a Quality Assurance Plan that conformed to an identified management system and generally complies with ISO 9001. The Plan addressed the organisation and management of the project, work procedures, environmental considerations, safety and risk control and test procedures.

The Plan also detailed the procedures to be used in verifying that the deliverables meet the required specification including:

- the procedures and methodologies to be used to verify that the deliverables meet the required specifications.
- Details of proposed calibration checks and methodology to be used to establish both

reference stations and ground test sites.

5.2.14 Post Survey Spatial Accuracy Report

Following successful capture of the aerial data a Post Survey Spatial Accuracy report was submitted and approved by Geoscience Australia.

Observed flight runs were provided in ESRI shape file format.

All reference and check points used to verify accuracy were provided in ESRI shape file format.

For analysis of ground comparisons refer to Section 3.0. Accuracy Assessment.

5.2.15 Metadata

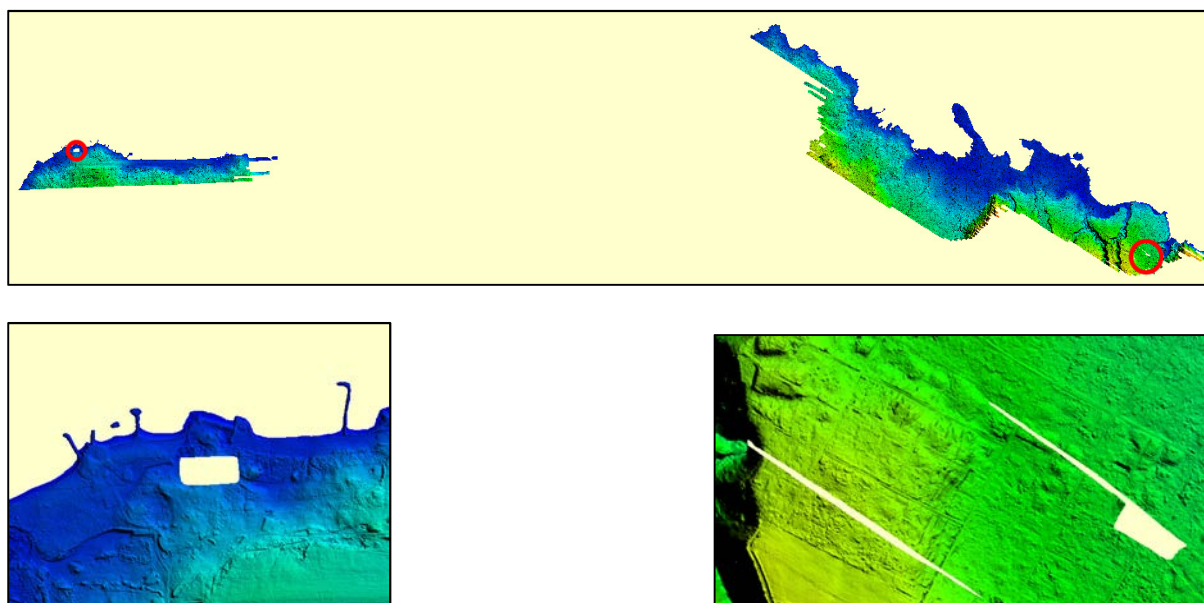
A complete metadata statement consistent with the current ANZLIC standard has been provided for each data product supplied. Metadata documents have been provided in Adobe Reader (.pdf) format. Further for every file in each data product, an .xml document with a complete metadata statement consistent with the current ANZLIC standard has been provided.

5.2.16 Data Coverage Problems and DEM Interpolation

A number of gaps in the data were identified in both the Apia and Airport regions of the survey. Geoscience Australia has requested that further DEMs and Contour products are generated with the data gaps interpolated.

ArcGIS 10 Terrains were created for each of topographic and bathymetric classified LiDAR datasets. Seamless ESRI terrains were used to form ESRI Grids derived from customised ArcGIS macros. A Natural Neighbours interpolation method has been employed to best triangulate between areas of greater point spacing.

The below images identify the location of the data gaps.



6 Spatial Accuracy

6.1 Uncertainty Model

Pelydryn has created a Total Propagated Uncertainty Model (TPU) for the HawkEye IIb that gives the uncertainty associated with each sounding, banded into a depth range. The TPU model takes individual sources of uncertainty from the various components of the survey system as well as environmental conditions and combines them into a series of values across depth bands using statistical RMS calculations.

These values are supplemented by the empirical data obtained from the factory and field calibration processes, the static navigation checks, the dynamic navigation checks and the settings selected by the Operator during data gathering operations.

Spatial data can be examined statistically to give an estimate of positional accuracy using the propagation of errors based around a Gaussian distribution. In this case, a linear distribution is considered and the standard deviation can be computed:

$$\sigma = \sqrt{\frac{\sum x^2}{n - 1}}$$

where: σ is one standard deviation of the linear error
 x is denoting the known residual errors of the process
 n is denoting the number of residual errors of the process

Figure 27: Calculation of Positional Accuracy

6.1.1 Navigation Solution

The HawkEye IIb is fitted with an Applanix POS AV 410 providing real time precision GNSS and inertial measurement. The accuracy of the POS AV system varies between 0.05m and 2m depending on the method used to obtained corrections. For this project, a post-processed navigation solution utilized POS PAC MMS processing software to give an accuracy of between 0.05m and 0.1m. Restricting aircraft turning rates, banking angles and speed changes further reduced the uncertainty associated with the navigation system. Operators monitored the PDOP of the real time navigation solution and used the Applanix INFUSION processing solution, supplemented where required with additional base stations. Further details of the Applanix POS AV specifications are at http://www.applanix.com/media/downloads/products/specs/POSAV_SPECS.pdf

6.1.1.1 Base Station

Corrections for the post processed GPS solution are obtained from either permanent reporting stations or base stations installed for the survey. The accuracy of the position of these stations introduces an element of uncertainty into the final position solution obtained. As the Base Station provides information on which corrections to be applied to the raw GPS gathered in the aircraft are based, the corrections can only ever be as good in absolute terms as the accuracy of the position of the base station itself.

During this survey, RINEX Corrections were obtained from stations specifically established for the survey. It is assessed these station accuracies were 0.03m for horizontal and 0.05m for the vertical.

6.1.2 HawkEye IIb System Uncertainty

The following HawkEye IIb system sources of uncertainty are factored in to the TPU model:

- HawkEye IIb Internal Offsets and Latency
- Scanner Position
- Laser Discrimination

6.1.2.1 Laser Footprint

Lasers are subject to beam divergence, which means that some beam spreading takes place in the air during airborne laser surveys. This gives rise to an altitude-dependant spot size at the sea surface, which is calculated using the design specified divergence for the altitude in use.

Light is spread further in the water column in the form of “Broadening”, giving rise to a broader Footprint of the laser on the seafloor. This Footprint on the seabed will be a function of the Spot size and the Broadening, which is itself a function of depth and the water clarity (k).

These two sources of uncertainty are combined to provide the positional uncertainty associated with the laser footprint at the seabed.

6.1.2.2 Surface Angle of Incidence

As light moves from one medium to another it is subject to refraction. In this case, refraction occurs as the light moves from the air into the water at the sea surface. The amount of refraction can be calculated using Snell’s Law (<http://www.opticalres.com/lt/illuminationfund.pdf>):

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Where

n_1 = the refractive index of Air (1.000292)

n_2 = the refractive index of Salt Water (1.3423)

θ_1 = the angle of incidence of the light with respect to the normal (90 – 10)

θ_2 = the refracted angle with respect to the normal

For the HawkEye IIb laser, the angle of incidence at the sea surface is 20 degrees which means that the refraction is 14 degrees. This is included in the position calculation of the centre of the laser footprint. However, this calculation assumes that the angle of the interface at the surface is horizontal, i.e. there is a flat sea surface. This is not always the case. If the sea surface has small waves or even ripples the actual angle at the sea surface can be different. And it will be different for each laser spot with some spots interfacing with the front of a wave, some interfacing with the top or bottom of a wave and some interfacing with the back of a wave.

Again using Snell’s Law, it can be seen that this angle of the sea surface at the point of interaction will give rise to different values of refraction.

When this happens there will be an uncertainty introduced to the horizontal position of the spot centre and also to the vertical value as the light is not now following the predicted path through the water column; it is taking a longer path and will hence give a deeper depth value than if it were following the route for a flat surface. The worst case occurs when the angle of interaction is at its greatest i.e. on the face of a wave.

The Hawkeye IIb is normally operated in sea states below sea state 3 although the actual angle of incidence is input based on conditions observed.

6.1.3 Tidal Observations and Co-Tidal Modelling

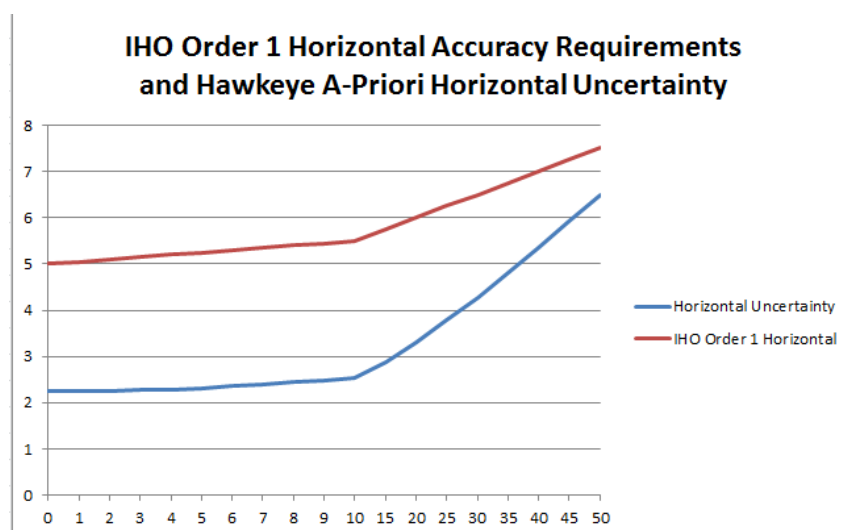
The TPU model also accepts values of uncertainty associated with the observance of tides and the use of co-tidal corrections or separation models. It is assumed that the derived separation model for this survey is accurate to 0.1m

6.1.4 Geoid Model and Transformations

The use of transformation calculations introduces further uncertainty to the final values. The HawkEye IIb operates in WGS84 as a default so any transformations required to meet the required output parameters may introduce uncertainty. This is quantified in the TPU model, but can be considered to be zero when using GRS80.

6.1.5 Empirical Observations

The data from the HawkEye IIb is validated against bathymetric data obtained by MBES and topographic data obtained by terrestrial survey observations or from higher order topographic data. A statistical comparison of the HawkEye IIb data is carried out against this control data and the results are used as an input to the uncertainty model. The accuracy and precision values specified for the higher order data are also input into the model, although it should be noted that the resultant uncertainty is worst case and as such the specified requirements are usually exceeded.



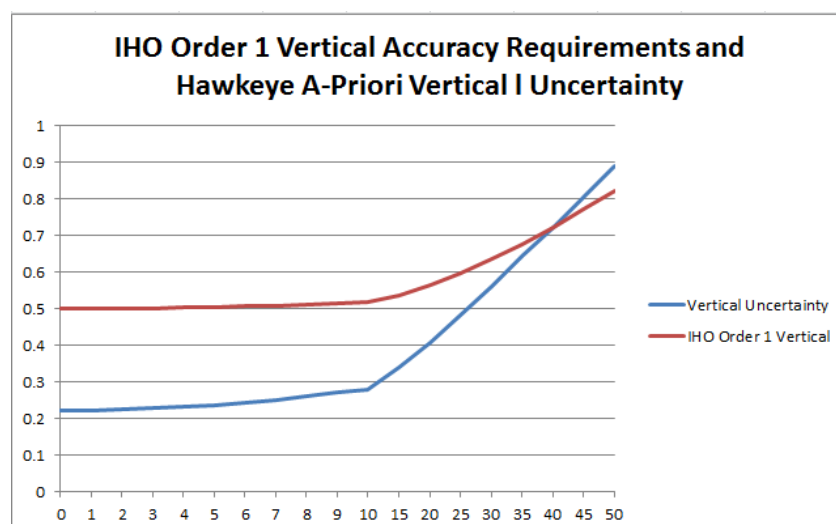


Figure 28: Horizontal and Vertical Horizontal Propagated Uncertainty and IHO Order 1 Required accuracies.

6.2 Accuracies Required for IHO Order 1

The International Hydrographic Organisation (IHO) has established standards to be met for Order 1 and detailed them in SP44 (5th Edition) as follows for vertical accuracy:

$$\pm \sqrt{a^2 + (b \times d)^2}$$

Where:

a represents that portion of the *uncertainty* that does not vary with depth

b is a coefficient which represents that portion of the *uncertainty* that varies with depth d is the depth

b x d represents that portion of the *uncertainty* that varies with depth

The vertical *uncertainty* at the 95% *confidence level* should be recorded

Horizontal accuracy is defined as 5m +/- 5% of the depth.

As an example for the depths experienced in the survey this equates to the following:

IHO Order 1 Horizontal Accuracy at 5m	5.25m
IHO Order 1 Vertical Accuracy at 5m	0.504m

6.3 Accuracies Achieved (95% Confidence)

Following examination of the acquired data it is inferred the following accuracies have been achieved in this survey:

LiDAR Error Budget	A Priori		Depth Range (m)				
	X/Y	Z	5	10	20	30	40
Tidal Measurements	0.05	0.05					
Co-Tidal model	0.00	0.00					
Base Station Position	0.05	0.05					
Navigation Solution GNSS + AV410 ¹	0.05	0.05					
HEII Internal Offsets and Latency ²	0.002	0.002					
HEII Lever Arms ³	0.005	0.005					
Scanner Position ⁴	0.001	0.001					
Sea Control Area ⁵	0.5	0.10					
Calibration Values ⁶	0.300	0.15					
Laser Discrimination ⁷	0	0.11					
Footprint Uncertainty x/y ⁸	2.200		2.256	2.417	2.973	3.720	4.565
Surface Angle of Incidence x/y	0		0.234	0.469	0.938	1.407	1.876
Surface Angle of Incidence z		0	0.063	0.125	0.251	0.376	0.501
Totals 	2.278	0.244					
A posteriori x/y			2.344	2.531	3.173	4.021	4.971
A posteriori z			0.255	0.285	0.381	0.502	0.634
Order 1 Vertical Requirements			0.504	0.516	0.596	0.634	0.721

Table 11: Horizontal and Vertical Accuracies Achieved

7 Survey Analysis

At all stages of the acquisition process, Pelydryn has undertaken a comprehensive analysis of the post processed navigational solution and the quality of the acquired bathymetric data. Having conducted Quality Assurance on the post processed navigation data it is assessed that the navigational solution has met the required IHO Horizontal and Vertical Positioning requirements.

Having confirmed the validity of the navigational solution, an analysis of the data coverage and density has been undertaken. The results for the density in the areas have been presented in Section 7.2. The density achieved utilising the acceptance criteria provided by Geoscience Australia has been exceeded in all areas that the HawkEye IIb System has obtained depth detections. However there are several observations to be made about the coverage and quality of the data that has been supplied.

7.1 Data Coverage

AAM/Pelydryn were contracted to undertake surveys on the polygons provided in the tender document. The entire extent of these polygons was flown once and it is assessed that the seabed was consistently detected at depths of up to 40m and in some instances in excess of 40m. The reef structures are complicated and provide unique challenges for hydrography. **Figure 299** below shows a typical reef structure:

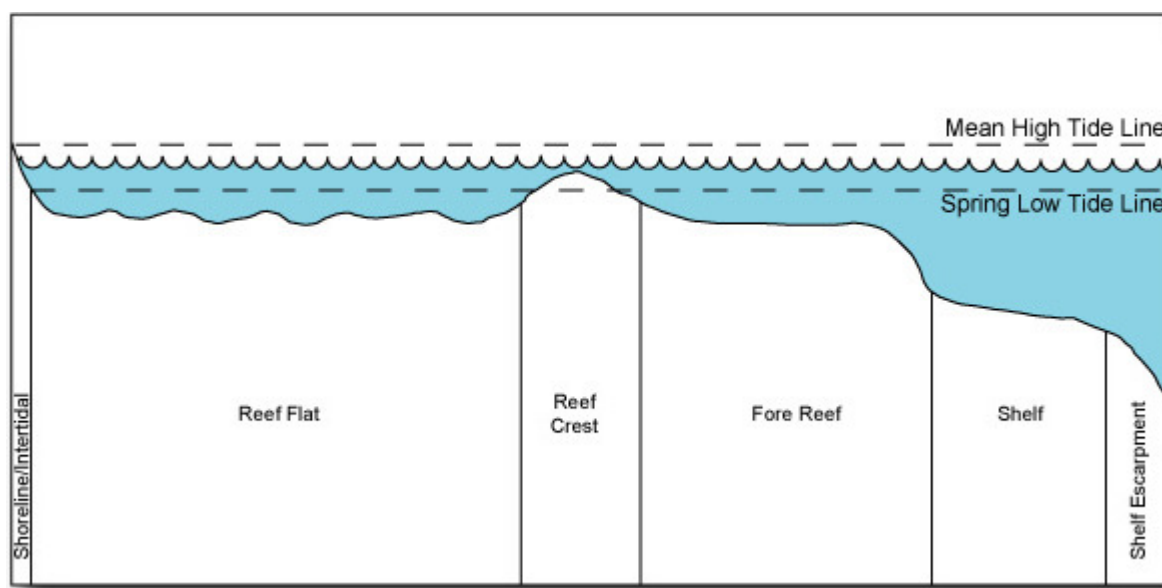
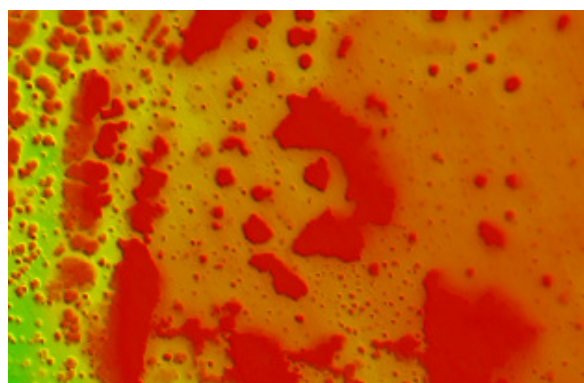


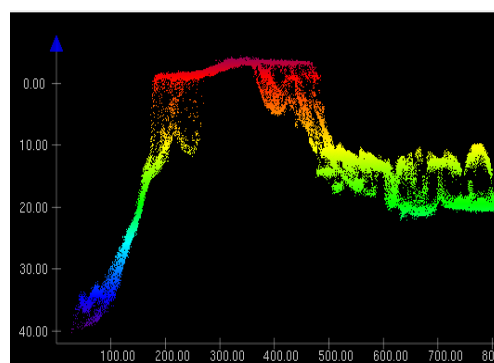
Figure 29: Typical Reef Structures

The reef flat is often in very shallow water and provides challenges to LiDAR systems because of the shallow depths encountered. Reef Crests are easily visible and often form small offshore islands. As the water gets deeper the fore reef and shelf can extend for significant distances before the reef edge and drop-off is encountered. Alternatively on the seaward side of fringing and patch reefs they can abruptly drop off in a near vertical wall. Within the reef flats and fore reef it is common to find isolated coral heads, gullies and highly irregular seabed due to the breakdown of dead coral to form boulders and new growth on and around it.

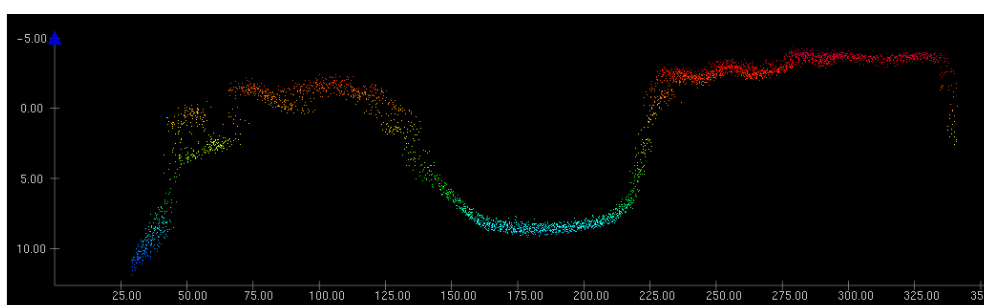
Typical Reef Flat or Fore Reef with isolated Coral Heads



Near Vertical Drop Offs



Typical Steep Sided Gullies



Gently Sloping Fore Reef

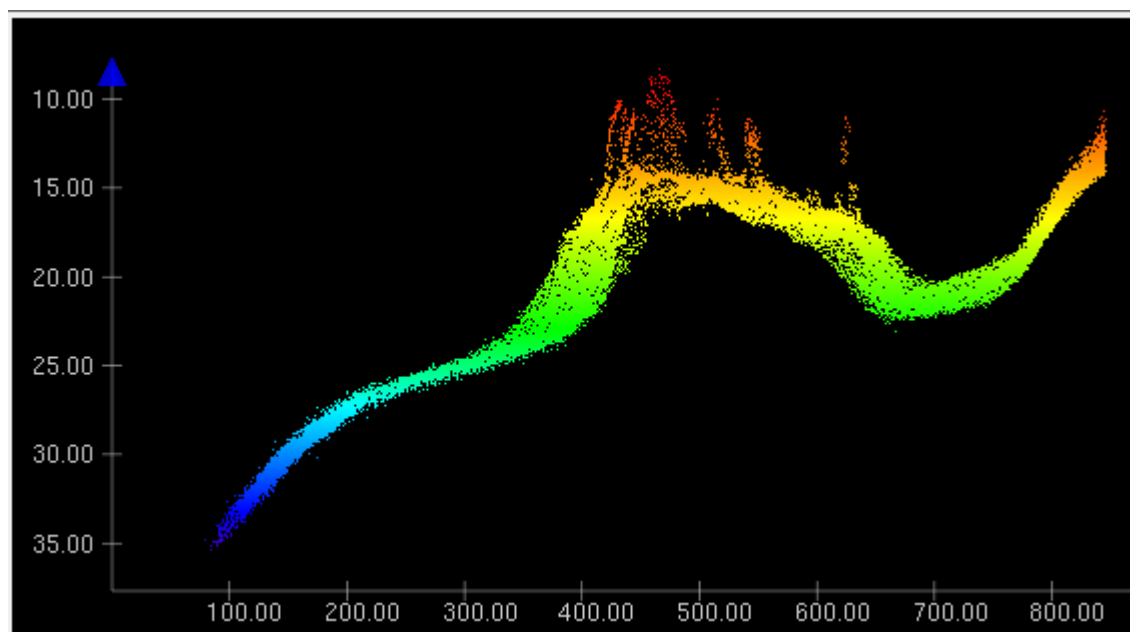


Figure 30: Typical Reef Topography encountered in Coastal Waters

The topography encountered varied significantly and when water clarity issues and steep slopes were encountered this would often reduce the depths observed. As a result, reflights were conducted in order to try to extend the coverage with variations in line direction used to try and

collect additional data. Where no additional coverage was found on a second flight the areas were designated “no bottom found” where no depths were encountered.

7.1.1 Coverage in Shallow Water

Paragraph 4.6.3 has documented the issues of operating the HawkEye IIb System in Shallow Water. Pelydryn undertook reflights in these areas in order to increase both coverage and density; however water clarity, the depth and the contrast of the seabed, has meant that coverage is not 100% in the very shallow regions. An example is shown in the image below (**Figure 31**) in Vaiusu Bay, where the combination of factors previously mentioned has affected the density of data gathered. It however has met the density requirements specified in the SoR.



Figure 31: Gaps in Data Coverage seen in Shallow Water (Data shown at 2m Resolution)

7.1.2 Coverage in Deep Water

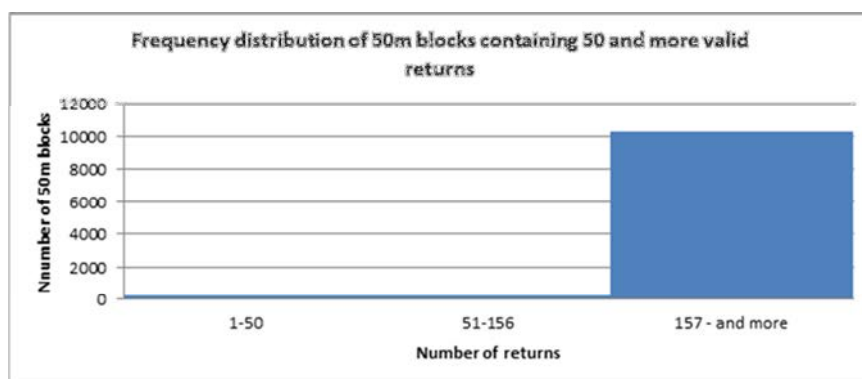
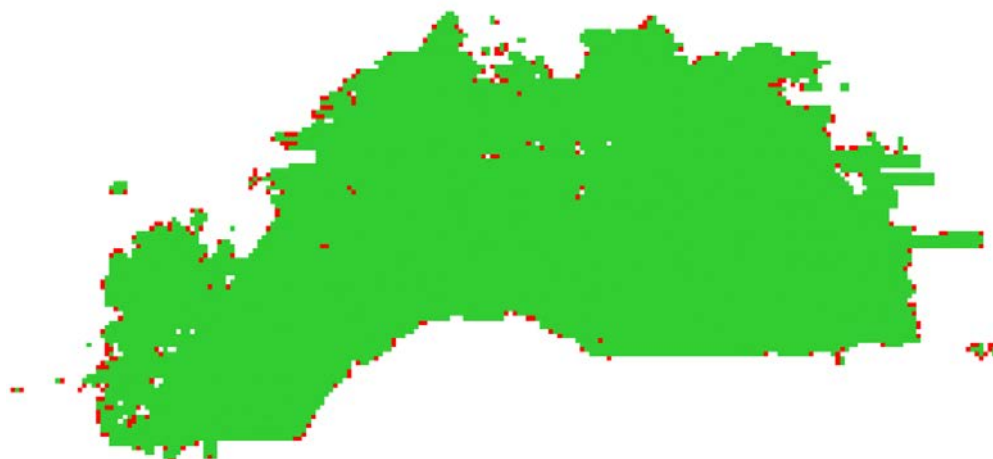
The offshore reefs provided a unique challenge to the HawkEye IIb System. Steep sided on the seaward face the reef plate was generally lying between 10m and in excess of 30m deep with isolated coral heads and pinnacles being much shallower or projecting above the water. Areas of poor coverage were reflighted with varying line directions to capture more data as required.

7.2 Survey Density

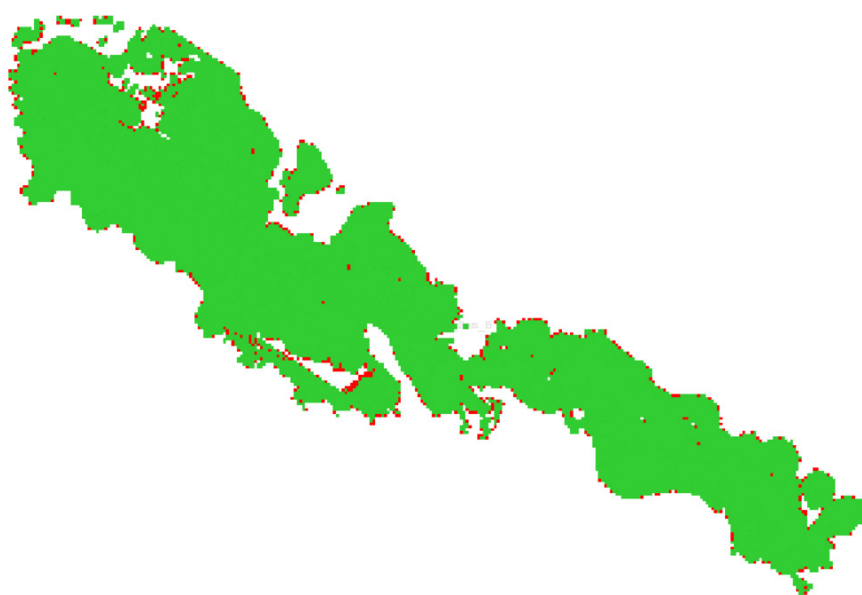
The environmental conditions experienced during the period of the survey, particularly the turbidity that resulted from the cycle of passing weather fronts and suspended sediments, meant that it was not possible to achieve the 100% coverage required in all areas during the period defined for the survey. Repeated coverage with different flightlines, sensor settings, wind conditions, tide heights and sun angle did not improve the seabed definition and where areas were flown twice with no detection it has been marked no bottom found.

To allow for the vagaries of the grid origin and orientation the final processed data was analysed at a grid cell size of 50m. Of this new grid, there are to be a minimum of 50 accepted data points or 156 for 100% Coverage in accordance with Section 11.5 of the Statement of Requirement.

Area A



Area B



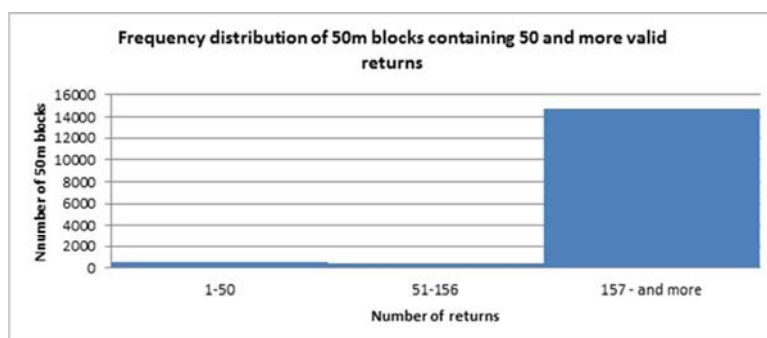


Figure 32: Final Density Results Block A and B

Apia Bathymetric Survey Density (85% of all 50m x 50m shall contain 50 or more soundings)	
Area A	97%
Area B	98%
Total Area	97.5%

Table 12: Densities achieved in Areas A and B.

The data collected meets the requirements of the contract and the data acceptance criteria provided by Geoscience Australia.

7.3 Topographic Integration Point Analysis

A number of sites lying adjacent to the area surveyed by the HawkEye IIb were co-ordinated and the surrounding area was surveyed using a RTK GPS.

The data from the observed sites was compared against the topographic data collected from lines which passed over the sites previously surveyed. Full details can be found at Annex F.

Apia A and B	TIP 1 Grass	TIP 1 Tarmac	TIP 2	TIP 3	TIP 4	Average
Average dz	0.002	0.023	-0.013	-0.13	-0.142	-0.052
Std deviation	0.059	0.08	0.098	0.074	0.061	0.0744

Table 13: TIP Analysis Results Apia A and B

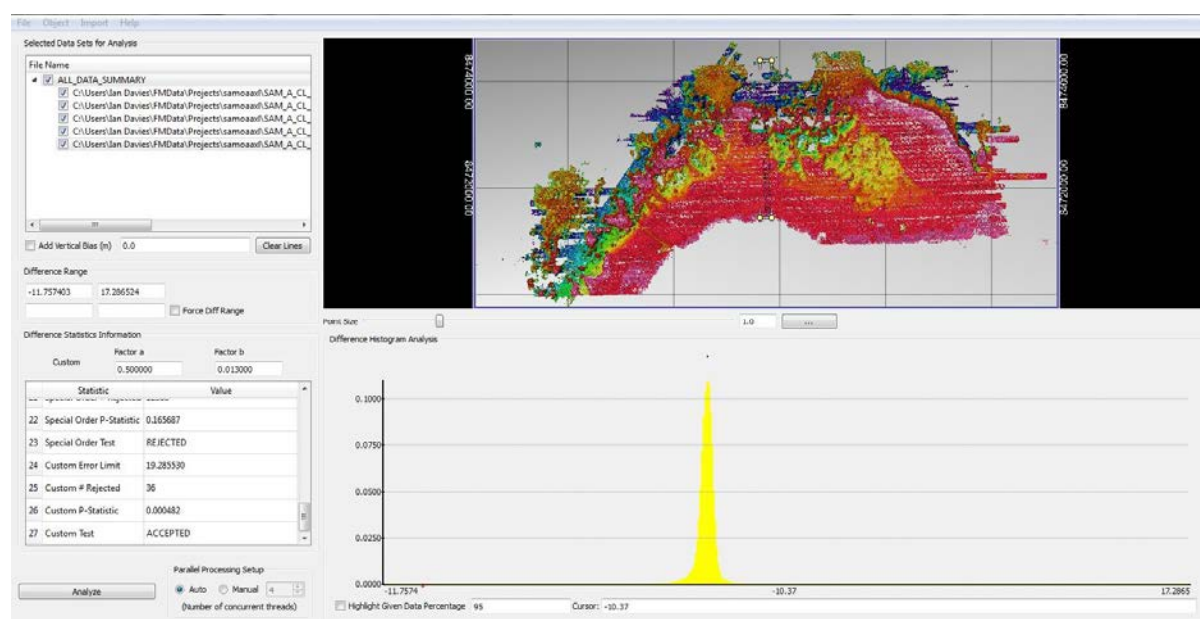
Analysis of the GNSS and Total Station EODM Observations show good agreement and low residual

errors, confirming the correct calibration and derivation of the point altitude from the topographic laser .

7.4 Cross-line Comparisons

Cross-lines were run perpendicular to the main sounding lines wherever possible. Analysis of the data quality was performed using the Crosscheck Tool within Fledermaus. At the core of this program is a point versus surface comparison tool, designed to generate difference statistics. The cross-line data was processed in CARIS HIPS, before being exported as an ASCII XYZ file. These XYZ files were then compared against the finalised Fledermaus Shallow Edited Surface. The difference statistics for each cross-line are detailed below, together with sample graphics for each survey Block.

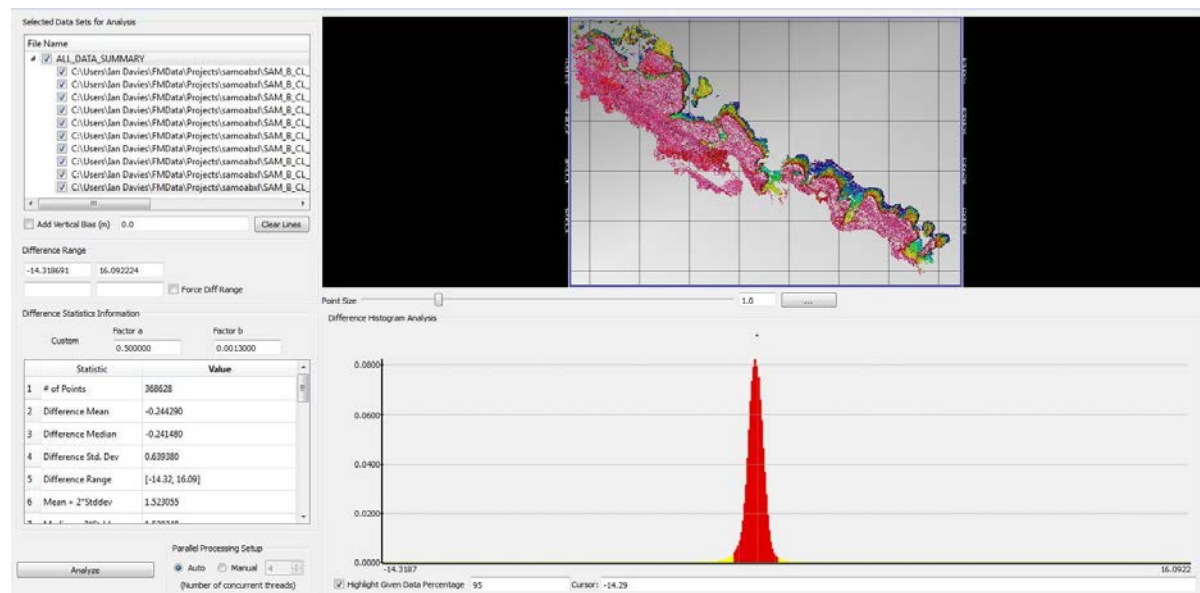
7.4.1 Hydro Crosscheck Results Block A



Area H Statistical Summary			
No Of Points	434667	Order 1 Error Limit	0.7091
Difference Mean	-0.0842	Order 1 Rejected	5314
Difference Median	-0.0798	Order 1 P- Statistic	0.062
Standard Deviation	0.424	Order 1 Test	ACCEPTED

Figure 33: Crosscheck Results Block A

7.4.2 Hydro Crosscheck Results Block B



Area B Statistical Summary			
No Of Points	368628	Order 1 Error Limit	0.709
Difference Mean	-0.2443	Order 1 Rejected	7200
Difference Median	-0.2414	Order 1 P-Statistic	0.0154
Standard Deviation	0.6393	Order 1 Test	ACCEPTED

Figure 34: Crosscheck Results Block B

The Cross-line Comparison analysis shows that the data has met IHO Order 1 specifications.

7.5 Survey Coverage and Comparison with the Bathymetric Chart

Whilst not a full charting survey, comparison was made between the survey data and Geotiffs of NZ Charts 864, 865 and 8655. The method used for comparisons was to resample and drape the Geotiff chart over the accepted DTM. As the system works by resampling the lower resolution to the higher resolution (chart to DTM) some distortion can occur. When comparing depths, the centre of gravity of the sounding has been taken as the position to be verified. A check is made to ensure the depth under the sounding footprint is similar and where different the general depth in the area is quoted to allow checking and verification. The advantage of draping and sampling the chart over the DTM with contour plots is to allow the surveyor to quickly see the area he needs to focus on (such as shoals) and any new features helping with the comparison process.

7.5.3 Area A

Lying between Paepaeala and Magia on the north western side of Upolo the area has depths ranging from drying heights to in excess of 35m when referred to Chart Datum at Apia. Area A is characterised by a wide heavily indented reef plate extending up to 3km from the shore with a steeply sloping reef plate of around 30 degrees at the drop off followed by a gentle slope of 1-2

degrees once the seabed is reached at 20-30m. A comparison with the bathymetric chart 864 shows good agreement with a number of small shoal features uncharted. Data coverage has been achieved to the project specification, with consistent depths of 30m (CD) achieved where the reef structures and turbidity allowed. Topographic data coverage of the coastline has been achieved and accompanies this report.

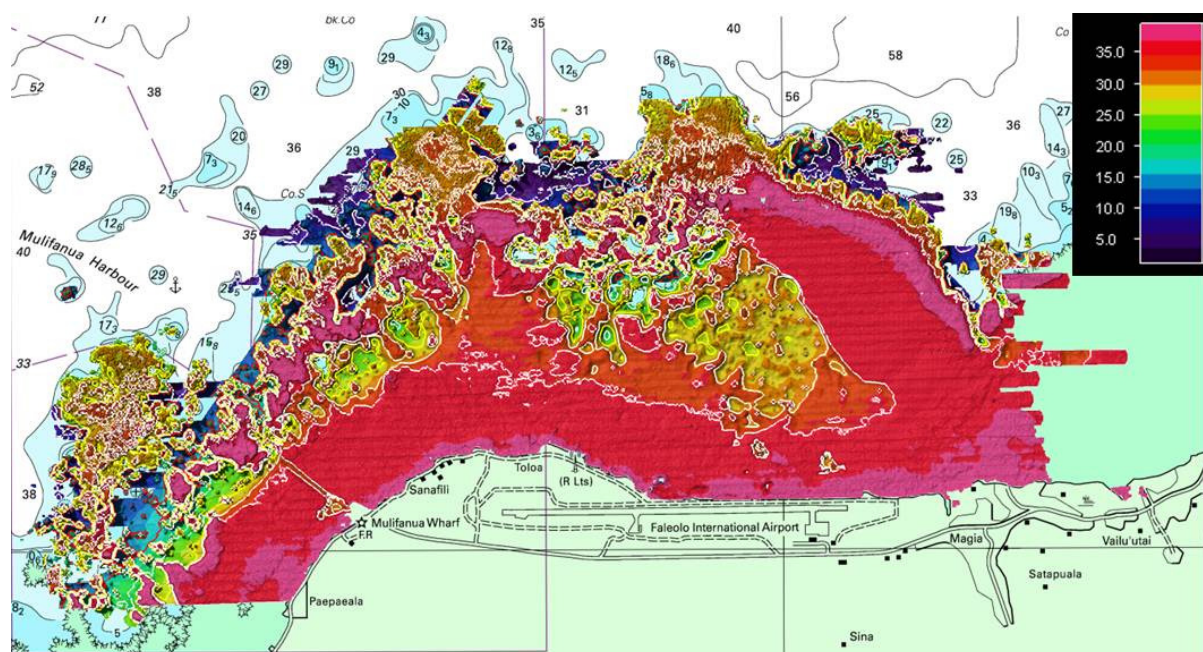


Figure 35: Area A gridded at 3m

One Hydrographic Note has been submitted with this report relating to uncharted shoals discovered during QA of the data:

H Note No	Remarks
H102-Pel-Samoa-001	Uncharted Wreck
H102-Pel-Samoa-009	Uncharted Shoals

Table 14: Hydrographic Notes raised for Area A

7.5.2 Area B

Area B lies to the East of Area A and covers the coast and hinterland around Apia between Faleula and Letogo. Examination of the bathymetric contours shows broad agreement with those charted with depths being in general agreement. The area is characterised by a highly indented reef plate whose average depth is less than 0.5m. A number of small deep bays exist along the coast bisecting the shallow areas with depths of 20m or more. The reef plate descends at an angle of up to 60 degrees in places and depths of 30m or more are encountered very close to the reef edge. A number of new shoals and contacts have been identified during QA and are submitted as Hydrographic Notes with this report. It is assessed that the required Bathymetric and Topographic data coverage of the area surveyed has been achieved

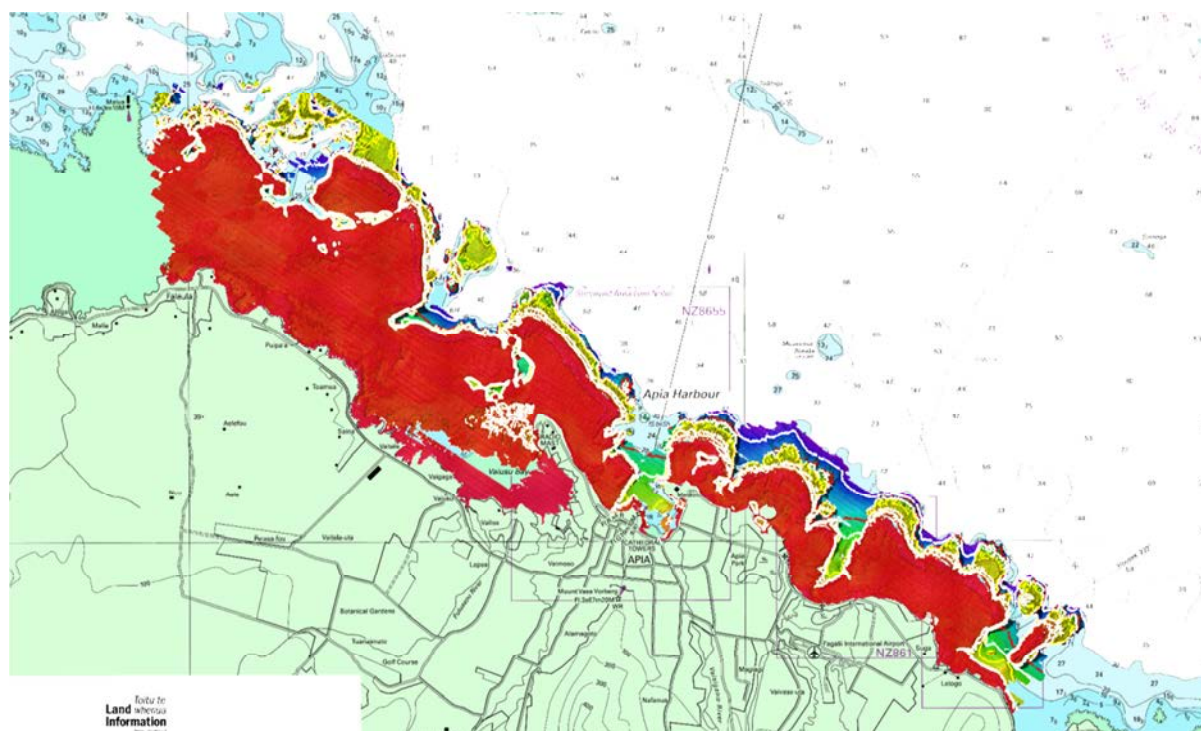


Figure 36: Area B gridded at 3m

H Note No	Remarks
H102-Pel-Samoa-002	Uncharted Wreck
H102-Pel-Samoa-003	Shoal Depth
H102-Pel-Samoa-004	Shoal Depth
H102-Pel-Samoa-005	Uncharted Stranded Wreck
H102-Pel-Samoa-006	Uncharted Wreck
H102-Pel-Samoa-007	Uncharted Shoal
H102-Pel-Samoa-008	Uncharted Shoal
H102-Pel-Samoa-010	Uncharted Shoals

Table 15: Hydrographic Notes raised for Area B

7.6 Wrecks/Contacts/Significant Features

The Field Processor examined all data in CSS initially as point clouds, and then in CARIS HIPS during field cleaning, utilising the 2D view to examine lines and the 3D view to examine areas. During this process the significant contacts or deviations from the adjacent soundings were noted. The plot of Standard Deviation in CARIS was also used to highlight areas of rapid changes in depth relative to the surrounding area.

Any significant features were examined more closely (including reviewing the wave forms from the feature) and a decision made on the probability of the return being valid and where necessary a Hydrographic Note was raised.

7.7 Vegetation Analysis

Due to insufficient land survey data in different vegetation classes on Samoa the Vanuatu data was used for investigation into the sensors capabilities. The Hawkeye IIB sensor remained unchanged between the Samoa and Vanuatu surveys and as such the results from the Hawkeye IIB data from Vanuatu were used for the analysis.

For the purposes of independent accuracy assessment check point measurements were taken throughout the Vanuatu project area. Check points were collected for each land cover type across the project extent. The Table below shows the number of check points for each class.

Area	Landcover Class	No. of Points
Vanuatu	Grass	38
	Grass/Bushes (1-5m)	10
	Grass Under Trees	33
	Grass Under Wooded Trees (10-12m)	7
	Low Shrubs/Grass Cover (0.5-1.0m)	55
	Sand Under Palms (4-10m)	25
	Shrubs (5m) under Trees (10m)	4

Comparison of the field check points with elevations interpolated from LiDAR data resulted in accuracies shown in the following tables for the landcover classes:

Vertical Accuracy – LiDAR Data					
Area	Landcover Class	No of Points	Mean Difference (m)	Std Deviation (m)	RMS (m)
Vanuatu	Grass	38	+0.030	0.087	0.091
	Grass/Bushes (1-5m)	10	-0.078	0.098	0.122
	Grass Under Trees	33	+0.068	0.071	0.097
	Grass Under Wooded Trees (10-12m)	7	-0.093	0.090	0.123
	Low Shrubs/Grass Cover (0.5-1.0m)	55	+0.010	0.084	0.084
	Sand Under Palms (4-10m)	25	+0.111	0.094	0.144
	Shrubs (5m) under Trees (10m)	4	-0.208	0.412	0.411

Vertical Accuracy Comparison with Data Specification – LiDAR Data (+/-0.30m)					
Area	Landcover Class	Within +/- 0.3m	Outside +/- 0.3m	% Within +/- 0.3m	% Outside +/- 0.3m
Vanuatu	Grass	38	0	100%	0%
	Grass/Bushes (1-5m)	10	0	100%	0%
	Grass Under Trees	33	0	100%	0%
	Grass Under Wooded Trees (10-12m)	7	0	100%	0%
	Low Shrubs/Grass Cover (0.5-1.0m)	55	0	100%	0%
	Sand Under Palms (4-10m)	24	1	95.8%	4.2%
	Shrubs (5m) under Trees (10m)	2	2	50%	50%

8 Conclusion

Within the operational limitations of the system, the HawkEye IIb has detected all significant shoals and features to the required 30m depth within the areas surveyed except where seabed slope, extreme shallow water or turbidity have reduced detection capabilities.

9 Deliverables

The following project deliverables have been Quality Assured and delivered:

Data Type	
Raw LiDAR Data	LAS Files by line containing unclassified Hydro and Topo Data
Raw GNSS Data	GNSS Base Station Files
Raw Imagery	Time stamped and sequentially numbered imagery by sortie SBET Trajectory File
Processed Data	ASCII XYZ + Classification of all accepted hydro and topo data in 1km blocks ASCII XYZ + Intensity in 1km Blocks Intensity at 5m resolution in ECW Format Ortho-rectified Imagery in ECW format DEM (bathymetric and topographic) DSM CHM FCM Contours (bathymetric and topographic) Seamless DEM and Contours.
Reporting	Report of Survey Daily Progress Reports Static Navigation Checks (PDF) POSPAC Processing Reports (PDF) Mobilisation Report Density Analysis in Shapefile Format Tidal Report (PDF) Shapefile Flight Trajectories Tidal Data

Annexes

Annex A Personnel

AAM Project Manager

Martin Tunwell

Pelydryn Project Manager

Ian Davies

Field Manager

Andrew Eastmond

Data Processing and QC

Julian Olszewski

Robin Fidler

Amy Davies

Almu Perez Gonzalez

James Plewman

Tom Waddington

Rebecca Carling

Sophie Nunn

Nurul Shaida Binti Nasir

Audi Hani Abdullah

System Operator

Rebecca Carling

Annex B Daily Progress Reports

Supplied as pdf.

Annex C Mobilisation Report

Supplied as pdf

Annex D POSPAC Processing Reports

Supplied as pdf

Annex E Static Navigation Checks

Supplied as pdf

Annex F Post Survey Spatial Accuracy Report (including tidal report)

Supplied as pdf